

Empirical Identification of a Variable-Structure Automaton under the P-Model

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Abstract

Learning-automata studies often require recovering the latent structure of a Variable-Structure Automaton (VSA)—policy, transitions, and penalty law—when only state–action–feedback interaction traces are observed. In a P-model environment the penalty probability is assumed to depend on the action but not on the source state; it remains unclear whether that assumption holds and whether estimates are stable under data loss. We address this setting with a fully empirical, count-based identification pipeline: validate the dataset, estimate normalized probabilities from sufficient statistics, identify an optimal action and ε -optimality gap, validate a forward simulator, stress-test estimates by random row removal, and audit per-state penalties $c_k(\phi_i)$ against the global P-model. The workflow is implemented as a reproducible end-to-end run on 3518 steps from 80 sequences. On the full dataset the lowest-penalty action is α_1 ($c_* \approx 0.20$), yet behavior is only weakly ε -optimal ($\varepsilon \approx 0.37$) and heterogeneous across states; a simulator matches empirical marginals within $L_1 \approx 0.016$. Under 20% subsampling (10 trials) the optimal action and penalty ranking are unchanged and ε moves little (0.371 ± 0.014), while per-state penalty spreads indicate moderate deviation from a strict action-only penalty law.

1 Introduction

Stochastic learning agents are frequently analyzed through Variable-Structure Automata (VSAs): finite-state controllers that select actions, update internal state, and receive binary environmental feedback. In the *P-model*, feedback $\beta \in \{0, 1\}$ indicates an unfavorable ($\beta = 1$) or favorable ($\beta = 0$) response, and a widely used identification target is the action-dependent penalty $c_k = P(\beta=1 \mid \alpha_k)$ together with the state–action policy $P(\alpha_k \mid \phi_i)$ and action-conditioned transitions $P(\phi_j \mid \phi_i, \alpha_k)$. A common working assumption is that c_k does not vary with the source state ϕ_i ; in practice, however, one must infer all of these quantities from observational interaction logs alone, assess whether behavior is nearly optimal with respect to the inferred penalties, and judge whether a generative simulator and a global penalty law are supported by the data.

This paper considers the inverse problem on a fixed interaction dataset with five states, three actions, 80 episodes, and 3518 recorded steps. Each observation records the sequence identifier, step index, source and destination states ($\phi_{\text{src}}, \phi_{\text{dst}}$), action α , and feedback β . Rather than fitting a parametric learner, we adopt a *nonparametric, count-based* scheme: empirical frequencies are accumulated, normalized where appropriate, and checked for internal consistency. The goals are to (i) validate the dataset and report sufficient statistics, (ii) estimate the VSA parameters and derived optimality quantities, (iii) compare simulated and empirical marginals, (iv) quantify sensitivity to random row removal, and (v) test the P-model via state-specific penalties $c_k(\phi_i)$.

The proposed workflow chains these stages in a single reproducible pipeline so that fingerprinting, estimation, ε -optimality analysis, simulation, perturbation, and model auditing share the same input and reporting conventions. Evaluation relies on explicit metrics— L_1 distances on marginals, the ε -optimality gap and spread of $P(\alpha^* \mid \phi_i)$, stability of α^* and ε under subsampling, and spread or drift of $c_k(\phi_i)$ relative to global c_k .

The main contributions are:

- An integrated empirical protocol that links validation, identification, generative checking, robustness testing, and critique of the P-model in one place.
- Explicit metrics at each stage so that results on the present dataset are comparable across runs and perturbation trials.
- A full empirical case study on the dataset above, including weak ε -optimality despite a clear low-penalty action, close agreement between simulated and empirical marginals, and stability of the optimal action under 20% row removal.
- Evidence that per-state penalty estimates exhibit non-trivial spread, so global c_k should be interpreted alongside $c_k(\phi_i)$ when assessing an action-only penalty law.

The remainder of this paper is organized as follows. Section 2 reviews the VSA/P-model setting and the count-based estimators. Section 3 describes the end-to-end identification pipeline and its implementation. Section 4 defines evaluation metrics, reports results, and discusses their interpretation. Section 5 summarizes the study. Appendix A provides supplementary source listings; Appendix B gives the complete experimental log.

2 Background

2.1 Variable-structure automata and the P-model

A **Variable-Structure Automaton** (VSA) is a stochastic decision model in which an agent visits discrete states ϕ_i , chooses actions α_k , and receives environment feedback. In this project the state set is $\{\phi_1, \dots, \phi_5\}$, the action set is $\{\alpha_1, \alpha_2, \alpha_3\}$, and each interaction step is recorded in the dataset together with a sequence identifier and step index. The learning goal is to recover, from many such sequences only, four objects required by the assignment:

- a **state-action policy** $P(\alpha_k \mid \phi_i)$,
- **transition dynamics** $P(\phi_j \mid \phi_i, \alpha_k)$,
- an **environmental penalty structure** (probabilities of unfavorable feedback), and
- a **norm of behavior** (e.g. how often the optimal action is taken in each state, via ε -optimality).

The environment is a **P-model**: feedback $\beta \in \{0, 1\}$ indicates a favorable (0) or unfavorable (1) response. The working assumption is that the *global* penalty $c_k = P(\beta=1 \mid \alpha_k)$ depends on the action but not explicitly on the source state—so one penalty vector suffices for Tasks 1–5. Task 6 examines whether that assumption is consistent with the data when penalties are estimated separately in each state.

2.2 Approach used in this project

The implementation is **purely data-driven**: every quantity is an empirical frequency or a ratio of counts from the provided CSV, followed by row-wise normalization where conditional distributions are required. We do not fit parametric models, run gradient-based learning, or impose structure beyond what the counts support. Consistency checks (policy rows and transition rows sum to one; global c_k matches the sample mean of β given α) validate the arithmetic.

The six assignment tasks form a pipeline: a **dataset fingerprint** (Section 2.3) validates scale and raw penalty rates; **estimation** builds counts and conditional probabilities; **optimal-action**

and ε -**optimality** analyses use the policy and global penalties; **simulation** draws synthetic trajectories from the learned model and compares marginals to the real data; **perturbation** randomly deletes rows and refits to test robustness; **model critique** compares per-state penalties $c_k(\phi_i)$ across ϕ_i to assess the P-model. Section 3 maps each step to code; Appendix A lists the corresponding excerpts.

2.3 Empirical estimators

All formulas below are evaluated on a (possibly subsampled) interaction table. Each CSV row is one step; we write $n \in R$ for row indices and S for the set of distinct `sequence_id` values. For a fixed sequence $s \in S$, J_s is how many rows belong to s (that sequence’s length in steps).

Dataset fingerprint. Before modeling, the assignment requires five summaries of the raw table:

- **Total number of rows** $|R|$ (one per step in the file).
- **Number of sequences** $|S|$ (distinct `sequence_id`).
- **Mean sequence length** $\bar{L} = \frac{1}{|S|} \sum_{s \in S} J_s$ (average of the per-sequence lengths J_s).
- **Empirical penalty rate per action** α_k : fraction of rows with $\alpha(n) = \alpha_k$ that have $\beta(n) = 1$,

$$\frac{1}{\#\{n \in R : \alpha(n) = \alpha_k\}} \sum_{\substack{n \in R \\ \alpha(n) = \alpha_k}} \beta(n).$$

- **Checksum** $C = \sum_{n \in R} \text{step}(n)$.

On the full dataset, each per-action rate above equals the global penalty c_k used in estimation (both are the sample mean of β given α_k).

Raw counts. The assignment requires these three tables *before* any probabilities are reported: they are the empirical basis for the policy $P(\alpha_k \mid \phi_i)$, the transitions $P(\phi_j \mid \phi_i, \alpha_k)$, and the penalties c_k . Each entry counts how many rows $n \in R$ match a pattern on `phi_src`, `phi_dst`, `alpha`, and `beta` (written $\Phi_{\text{src}}(n)$, $\Phi_{\text{dst}}(n)$, $\alpha(n)$, $\beta(n)$ below).

$N(\phi_i, \alpha_k)$ — **actions taken in a state.**

$$N(\phi_i, \alpha_k) = \#\{n \in R : \Phi_{\text{src}}(n) = \phi_i, \alpha(n) = \alpha_k\}. \quad (1)$$

$N(\phi_i, \alpha_k, \phi_j)$ — **transitions under an action.**

$$N(\phi_i, \alpha_k, \phi_j) = \#\{n \in R : \Phi_{\text{src}}(n) = \phi_i, \alpha(n) = \alpha_k, \Phi_{\text{dst}}(n) = \phi_j\}. \quad (2)$$

$N(\alpha_k, \beta)$ — **environment feedback per action.**

$$N(\alpha_k, \beta) = \#\{n \in R : \alpha(n) = \alpha_k, \beta(n) = \beta\}. \quad (3)$$

Write $r_i = \sum_k N(\phi_i, \alpha_k)$ for the row total over actions in state ϕ_i .

Conditional estimates (Tasks 1–4). For visited states ($r_i > 0$),

$$P(\alpha_k \mid \phi_i) = \frac{N(\phi_i, \alpha_k)}{r_i}. \quad (4)$$

For each action α_k , transition probabilities normalize each row of $N(\phi_i, \alpha_k, \phi_j)$ over ϕ_j (empty rows yield zeros). The **global** P-model penalty is

$$c_k = P(\beta=1 \mid \alpha_k) = \frac{N(\alpha_k, 1)}{N(\alpha_k, 0) + N(\alpha_k, 1)}. \quad (5)$$

Task 2 sets $\alpha^* \in \arg \min_k c_k$. Task 3 uses visited states $\mathcal{V} = \{\phi_i : r_i > 0\}$ and $\varepsilon = 1 - \min_{\phi_i \in \mathcal{V}} P(\alpha^* \mid \phi_i)$. Task 4 samples $\alpha \sim P(\cdot \mid \phi)$, $\phi' \sim P(\cdot \mid \phi, \alpha)$, and $\beta \sim \text{Bernoulli}(c_\alpha)$ along each real sequence length.

Per-state penalties (Task 6). To test action-only penalties, for each pair (ϕ_i, α_k) with $N(\phi_i, \alpha_k) > 0$,

$$c_k(\phi_i) = P(\beta=1 \mid \alpha_k, \phi_i) = \frac{N(\phi_i, \alpha_k, \beta=1)}{N(\phi_i, \alpha_k)}, \quad (6)$$

where $N(\phi_i, \alpha_k, \beta=1)$ counts penalized steps with that state and action. Under a strict P-model, $c_k(\phi_i)$ should vary little in ϕ_i for fixed k ; a diagnostic is $\max_i c_k(\phi_i) - \min_i c_k(\phi_i)$ per action.

3 Methodology

The implementation comprises one module per task and is run end-to-end via `python main.py` $\rightarrow$ `load_all` $\rightarrow$ `run_all`. Formulas are in Section 2.3; Appendix A lists the matching source.

3.1 General flow

The end-to-end pipeline is implemented in `main.run_all` (Appendix A.6). It always validates the raw CSV, fits the VSA quantities on the full dataset, and optionally repeats the fit on random row subsamples to test stability. Per-state penalty tables are printed once at the end so the console log stays readable. Figure 1 summarizes `run_all` with decisions and loops; Algorithm 1 states the same logic in pseudocode.

3.2 Implementation details

The subsections below describe how each module realizes one part of the methodology. For each file we state the goal, point to the matching appendix listing, and list the main third-party libraries used (the standard library supplies `pathlib`, `dataclasses`, `argparse`, and `typing` where needed).

3.2.1 Software map

Table 1 lists each project module, its main entry point(s), return type, and the matching appendix listing (file by file, no task labels).

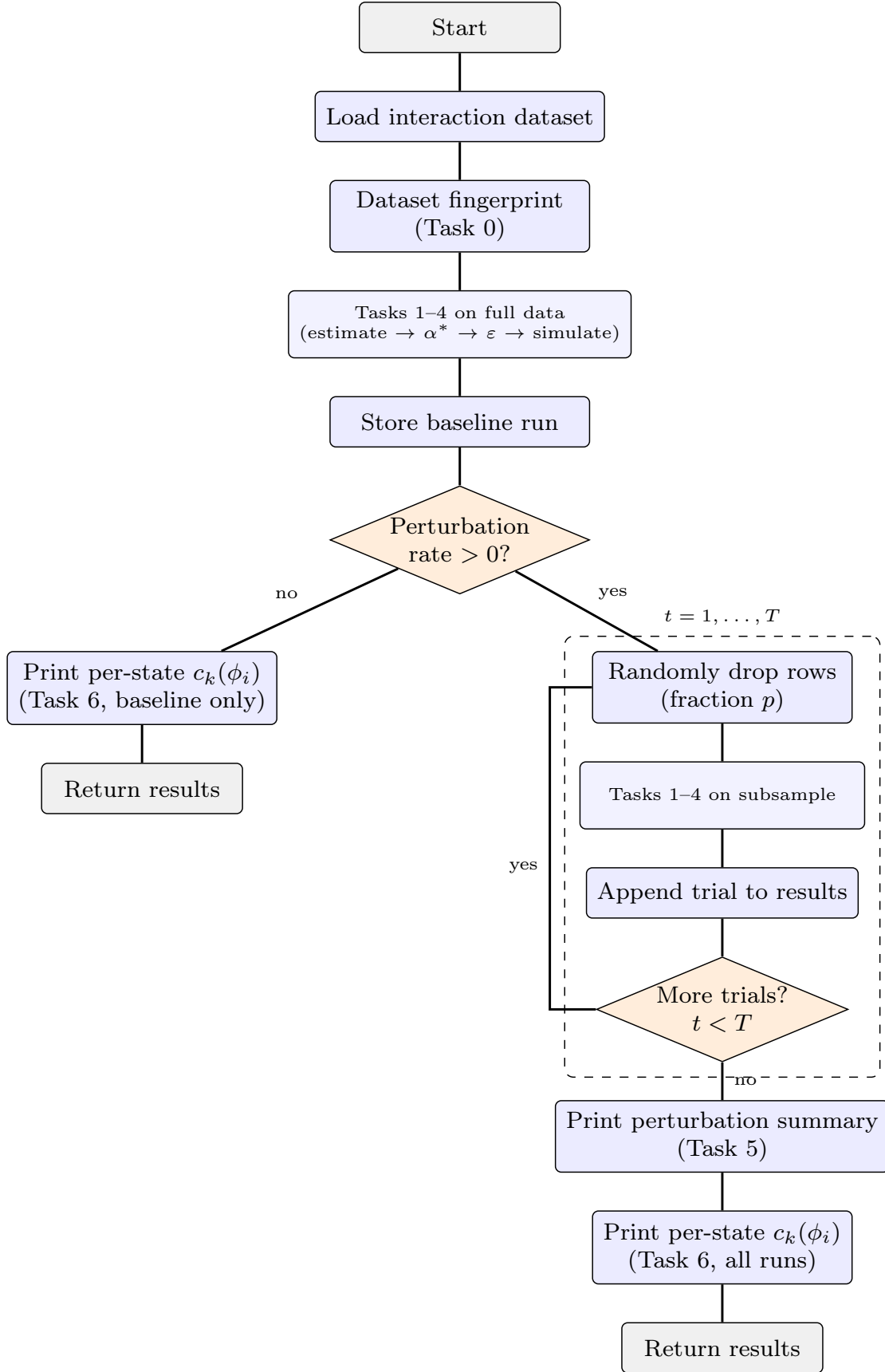


Figure 1: Flowchart for `run_all`: baseline Tasks 1-4 on the full dataset, optional perturbation loop over T trials (random row removal, refit, append), then Task 5 aggregate reporting when perturbation is enabled and Task 6 per-state penalty tables (baseline only if $p = 0$, else all stored runs).

Algorithm 1 General flow (`main.run_all`)

Require: CSV path, `perturbation_rate`, `experiment_count`, `random_seed`

```
1:  $df_{full} \leftarrow \text{load\_vsa\_dataset}(\text{path})$  ▷ util.py
2: run_fingerprint(path) ▷  $|R|$ ,  $|S|$ ,  $\bar{L}$ , penalty rates,  $C$ 
3:  $baseline \leftarrow \text{RUNTASKS1TO4}(df_{full}, baseline)$ 
4:  $results \leftarrow [baseline]$ 
5: if perturbation_rate > 0 then
6:   for  $t = 1$  to experiment_count do
7:      $df_p \leftarrow \text{\_drop\_rows\_randomly}(df_{full}, \text{rate}, \text{rng})$ 
8:     append RUNTASKS1TO4( $df_p$ ,  $t$ ) to  $results$ 
9:   end for
10: \_print\_experiments\_summary(baseline, perturbed runs)
11: end if
12: \_print\_c\_k\_per\_state\_runs(results)

13: function RUNTASKS1TO4(df, label)
14:    $est \leftarrow \text{run\_estimation}(df, \text{print\_penalty\_per\_state}=\text{False})$ 
15:    $opt \leftarrow \text{run\_optimal\_action}(est)$ 
16:    $eps \leftarrow \text{run\_epsilon\_optimality}(est, opt)$ 
17:    $sim \leftarrow \text{run\_simulation}(est, \text{path})$ 
18:   return PipelineRunResult(label, est, opt, eps, sim)
19: end function
```

Table 1: Source modules in the Python implementation, as used by `main.run_all` via `python main.py`.

Module	Entry point(s)	Return type	Ref.
<code>fingerprint.py</code>	<code>run_fingerprint</code>	Fingerprint-Result	A.1
<code>estimation.py</code>	<code>run_estimation</code>	Estimation-Result	A.2
<code>optimal_action_identification.py</code>	<code>run_optimal_action</code>	OptimalAction-Result	A.3
<code>e_optimality_analysis.py</code>	<code>run_epsilon_optimality</code>	Epsilon-Optimality-Result	A.4
<code>simulation.py</code>	<code>run_simulation</code>	Simulation-Result	A.5
<code>main.py</code>	<code>run_all</code> , <code>load_all</code>	PipelineRun-Result (list)	A.6
<code>util.py</code>	<code>load_vsa_dataset</code>	DataFrame	A.7

3.2.2 Dataset fingerprint (`fingerprint.py`)

Goal. Before modeling, report $|R|$, $|S|$, $\bar{L}$, empirical penalty rate per action, and checksum C (Section 2.3) so the submission can be validated against the assignment fingerprint.

Code. Appendix A.1 (`run_fingerprint`, lines 86–99), invoked at the start of Algorithm 1.

Libraries. `pandas` (`groupby`, `mean`, `sum`).

3.2.3 Estimation (`estimation.py`)

Goal. From the interaction table, build raw count matrices and derive $P(\alpha_k \mid \phi_i)$, $P(\phi_j \mid \phi_i, \alpha_k)$, global c_k , and per-state $c_k(\phi_i)$; run consistency checks (Section 2.3).

Code. Appendix A.2 (`run_estimation`, lines 104–189). From `main`, per-state penalties are computed but not printed (`print_penalty_per_state=False`). Optional CSV output is written under `outputs/task1_estimation/`.

Libraries. `pandas` (`groupby`, `unstack`, `normalization`); `NumPy` (`allclose` for checks).

3.2.4 Optimal action (`optimal_action_identification.py`)

Goal. Identify $\alpha^* \in \arg \min_k c_k$ from the penalty vector and record ties deterministically.

Code. Appendix A.3 (`run_optimal_action`, lines 109–128); uses `estimation.penalty` in memory or `penalty.csv` on disk.

Libraries. `pandas` (sorting, framing c_k); `NumPy` (`isclose` for tie detection).

3.2.5 ε -optimality (`e_optimality_analysis.py`)

Goal. Compute $\varepsilon = 1 - \min_{\phi_i \in \mathcal{V}} P(\alpha^* \mid \phi_i)$ over visited states only.

Code. Appendix A.4 (`_compute_core` / `run_epsilon_optimality`, lines 110–130).

Libraries. `pandas` (policy and count matrices); `NumPy` (`min`, `isclose`).

3.2.6 Simulation (`simulation.py`)

Goal. Generate synthetic trajectories with the learned policy, transitions, and Bernoulli penalties; compare data vs. simulated marginals and penalty rates.

Code. Appendix A.5 (`_simulate_block`, `run_simulation`, lines 134–184).

Libraries. `pandas` (marginals, alignment); `NumPy` (`default_rng`, sampling, binomial draws).

3.2.7 Perturbation experiment (`main.py`)

Goal. Delete a random fraction of rows, refit the model on each subsample, and compare α^* , ε , and simulation L1 to the baseline (Algorithm 1, perturbation branch).

Code. Appendix A.6: `_drop_rows_randomly` (lines 36–51), `run_all` perturbation loop (lines 319–344), `_print_experiments_summary` (lines 162–255).

Libraries. `NumPy` (`default_rng`, choice for row indices); `pandas` (`drop`, `reset_index`).

3.2.8 Model assumption check (`estimation.py`, `main.py`)

Goal. Form $c_k(\phi_i)$ for each state–action pair and compare across states (spread diagnostic); optionally across perturbation trials (Section 2.3).

Code. `penalty_per_state` in Appendix A.2; printing and spread helpers in Appendix A.6 (`_spread_across_states`, `_print_c_k_per_state_runs`, lines 70–78, 150–159).

Libraries. `pandas` (per-state table); `NumPy` (max/min spread, absolute differences). Also see `_run_tasks_1_to_4` and `run_all` in Appendix A.6 (lines 91–133, 302–346).

3.2.9 Shared data loading (`util.py`)

Goal. Load the headerless interaction CSV into a typed `pandas.DataFrame` and define shared result containers passed between modules.

Code. Appendix A.7: column constants, `EstimationResult`, `OptimalActionResult`, and `load_vsa_dataset`.

Libraries. `pandas` (`read_csv`, frame types).

4 Experiments

All numbers below come from a single orchestrated run (`pythonmain.py` via `load_all`: full dataset, 20% random row removal, 10 perturbation trials, `random_seed=42`). The complete console log is in Appendix B.

4.1 Evaluation metrics and methods

Dataset and protocol. Input: course dataset `Engin-vsa_dataset_full.csv` ($|R| = 3518$ steps, $|S| = 80$ sequences, mean length $\bar{L} = 43.98$). Each run loads the CSV once, prints the dataset fingerprint (Task 0), fits Tasks 1–4 on the full table (baseline), then optionally repeats Tasks 1–4 on ten subsamples with 704 rows ($\approx 20\%$) removed uniformly at random without replacement.

Metrics by task.

- **Fingerprint:** $|R|$, $|S|$, $\bar{L}$, empirical $P(\beta=1 \mid \alpha_k)$ per action, checksum $C = \sum_n \text{step}$.
- **Task 1 (estimation):** raw counts $N(\phi_i, \alpha_k)$, $N(\phi_i, \alpha_k, \phi_j)$, $N(\alpha_k, \beta)$; derived $P(\alpha_k \mid \phi_i)$, $P(\phi_j \mid \phi_i, \alpha_k)$, global $c_k = P(\beta=1 \mid \alpha_k)$; normalization checks (Section 2.3).
- **Task 2 (optimal action):** $\alpha^* \in \arg \min_k c_k$ and $c_* = \min_k c_k$.
- **Task 3 (ε -optimality):** $\varepsilon = 1 - \min_{\phi_i \in \mathcal{V}} P(\alpha^* \mid \phi_i)$ over visited states; spread $= \max_i P(\alpha^* \mid \phi_i) - \min_i P(\alpha^* \mid \phi_i)$.
- **Task 4 (simulation):** synthetic trajectories with the learned policy, transitions, and $\beta \sim \text{Bernoulli}(c_\alpha)$; report L_1 gaps between data and simulated marginals (actions, ϕ_{src} , ϕ_{dst}) and $|\bar{\beta}_{\text{data}} - \bar{\beta}_{\text{sim}}|$ per action.
- **Task 5 (perturbation):** stability of α^* , ε , and simulation L_1 (actions) across trials; mean/std of ε vs. baseline.
- **Task 6 (model check):** per-state penalties $c_k(\phi_i)$; spread $\max_{\phi_i} c_k(\phi_i) - \min_{\phi_i} c_k(\phi_i)$ per action; maximum absolute cell change $|\hat{c}_k(\phi_i) - \hat{c}_k^{\text{base}}(\phi_i)|$ vs. baseline.

4.2 Experimental results

Baseline (full data). Fingerprint checksum $C = 83035$ matches the assignment reference. Global penalties are ordered $c_1 < c_2 < c_3$ with $c_1 = 0.200$, $c_2 = 0.382$, $c_3 = 0.518$, so $\alpha^* = \alpha_1$ and $c_* = 0.200$. The empirical policy is biased toward α_1 in every state ($P(\alpha_1 \mid \phi_i) \in [0.63, 0.83]$), yet $\varepsilon = 0.370$ because ϕ_5 still chooses α_1 only about 63% of the time—behavior is *weakly* optimal and heterogeneous across states (spread 0.201). Simulation tracks the data well: $L_1 = 0.016$ on actions and $|P(\beta=1)_{\text{data}} - P(\beta=1)_{\text{sim}}| = 0.0077$ globally.

Perturbation (20% row removal, 10 trials). $\alpha^* = \alpha_1$ in *all* trials. Perturbed ε averages 0.371 ± 0.014 (only $+0.0013$ vs. baseline); individual trials span $[0.351, 0.394]$. Action-marginal simulation error stays small ($L_1^{\text{act}} \in [0.0028, 0.0210]$, baseline 0.0159). Task 6 spreads grow slightly under subsampling (e.g. α_2 : baseline spread 0.100 vs. perturbed mean 0.121), and the largest per-cell shift in $c_k(\phi_i)$ reaches 0.094 (mean 0.052 over trials).

4.3 Experimental details

Tables 2–3 collect the main numerics; Figures 2–5 visualize them. Per-trial count matrices and full Task 1–4 printouts are omitted here (Appendix B).

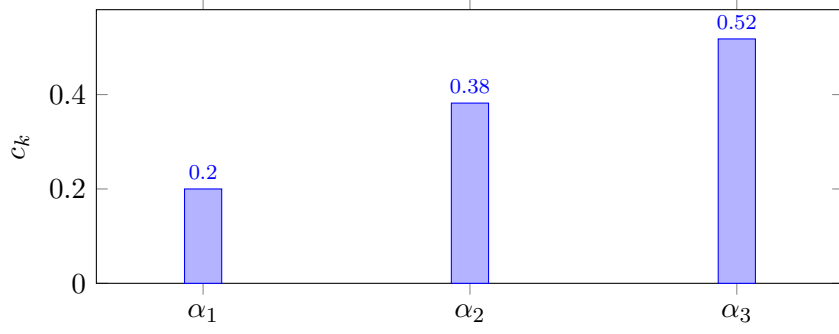


Figure 2: Global penalty probabilities c_k (baseline).

Figure 2 gives the global penalty ordering $c_1 < c_2 < c_3$ (0.20, 0.38, 0.52), so $\alpha^* = \alpha_1$ with $c_* \approx 0.20$. The separation between actions is clear enough that the optimal-action label is not sensitive to small count noise on this sample.

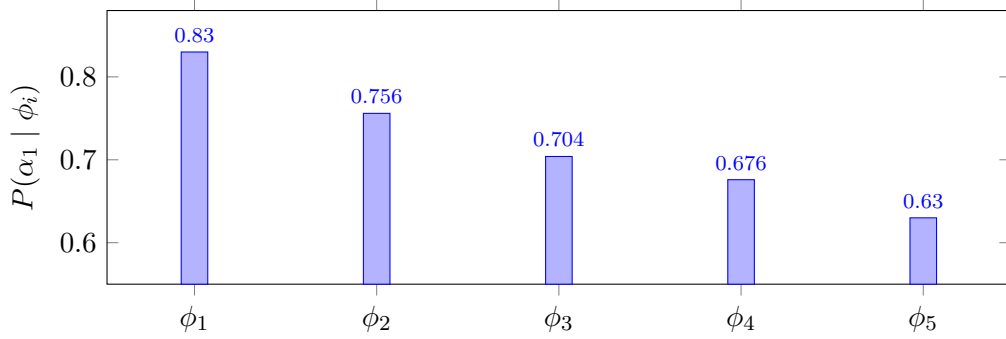


Figure 3: Empirical $P(\alpha^* | \phi_i)$ on visited states (baseline, $\alpha^* = \alpha_1$). State ϕ_5 attains the minimum and sets ε .

Figure 3 shows that $P(\alpha_1 | \phi_i)$ decreases from ϕ_1 to ϕ_5 (about 0.83 down to 0.63). State ϕ_5 therefore sets $\varepsilon \approx 0.37$ even though every state favors α_1 : behavior is weakly ε -optimal with a cross-state spread of 0.20 in $P(\alpha^* | \phi_i)$.

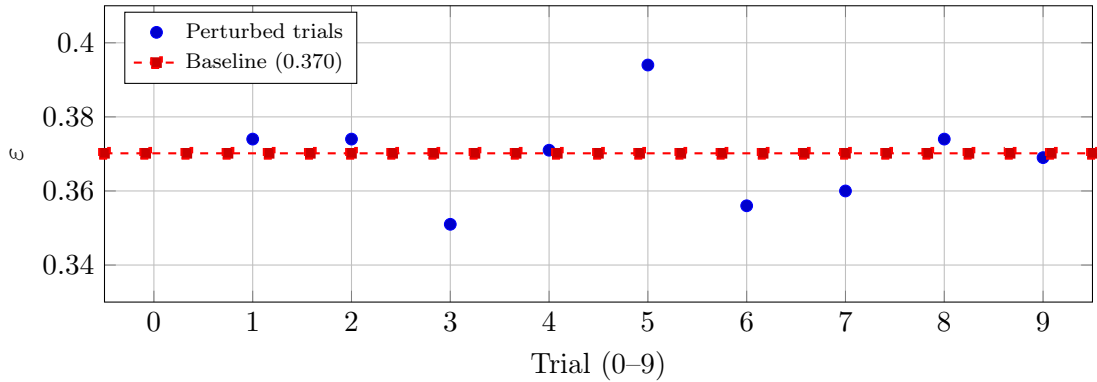


Figure 4: ε across perturbation trials vs. baseline (Task 3).

Figure 4 plots ε for ten perturbation trials against the baseline dashed line at 0.370. All points lie in $[0.351, 0.394]$ with mean 0.371 ± 0.014 , so subsampling shifts optimality gap little even though α^* stays α_1 in every trial (Table 3).

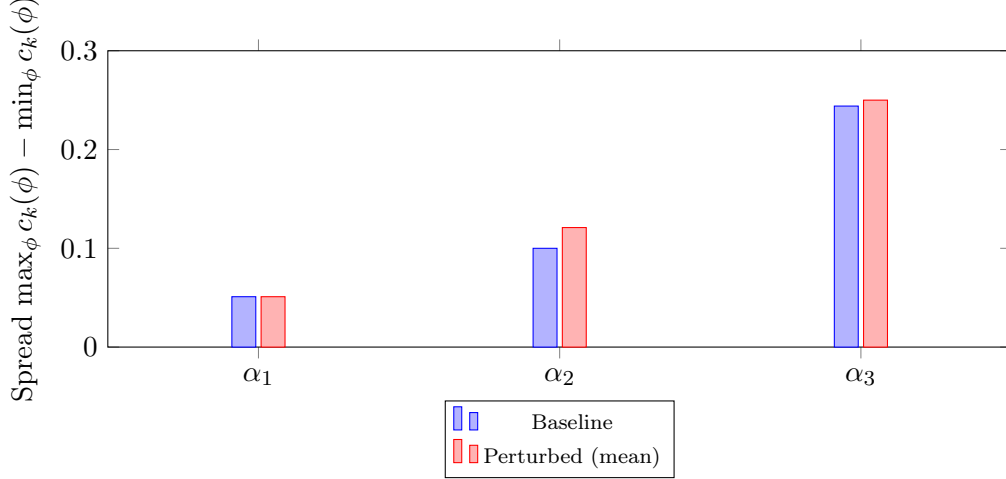


Figure 5: Task 6 spread of $c_k(\phi_i)$ across states: baseline vs. mean over perturbed trials.

Figure 5 compares Task 6 spreads of $c_k(\phi_i)$ across states. At baseline, α_3 has the largest spread (0.244); perturbed means rise for α_2 and α_3 (0.121 and 0.250), while α_1 stays near 0.051. Together with a mean maximum cell drift of 0.052 (max 0.094 in Table 3), the plots indicate measurable state dependence in penalties beyond a single global vector per action.

Table 2: Baseline metrics on the full dataset ($n = 3518$ rows).

Quantity	Value
$ S , \bar{L}, \text{checksum } C$	80, 43.98, 83035
c_1, c_2, c_3	0.200, 0.382, 0.518
α^*, c_*	$\alpha_1, 0.200$
$\varepsilon, \min_i P(\alpha^* \phi_i)$	0.370, 0.630 (state ϕ_5)
L_1 (actions / ϕ_{src} / ϕ_{dst})	0.016 / 0.015 / 0.016
Task 6 spread ($\alpha_1, \alpha_2, \alpha_3$)	0.051, 0.100, 0.244

Table 3: Perturbation summary (10 trials, 704 rows removed each).

Metric	Result
α^* (all trials)	α_1
ε : mean $\pm$ std (vs. baseline 0.370)	0.371 ± 0.014
L_1^{act} range	[0.0028, 0.0210]
Task 6 spread mean ($\alpha_1, \alpha_2, \alpha_3$)	0.051, 0.121, 0.250
$ \Delta c_k(\phi_i) $ max / mean over trials	0.094 / 0.052

Table 2 records baseline simulation L_1 gaps near 0.016 on actions and states and a global penalty-rate mismatch of 0.008, so the fitted policy, transitions, and Bernoulli penalties are mutually consistent with the observed histograms on the same file. Table 3 confirms that L_1^{act} under perturbation stays below 0.021 while α^* and the $\{c_k\}$ ordering are unchanged.

4.4 Discussion

Taken together, the figures and tables describe a VSA that is easy to summarize globally but not uniformly optimal in state space. A single low-penalty action (Figure 2) coexists with substantial slippage in $P(\alpha^* | \phi_i)$ (Figure 3), so reporting only c_* would overstate how closely the controller

follows α^* . The perturbation panel (Figure 4) and summary table show that this picture is stable under random row loss on the present sample, while the spread chart (Figure 5) warns against reading a strict action-only penalty law into the data. Simulation agreement in Table 2 supports internal coherence of the estimated model as a descriptive fit, not as proof of a true generative process.

The analysis is limited to one log, non-smoothed counts, and i.i.d. row deletion that ignores episode structure; rare (ϕ_i, α_k) cells may still drive part of the spread in $c_k(\phi_i)$. Global c_k remain useful for ranking actions and match per-state tables on α^* , but $c_k(\phi_i)$ should be reported whenever a P-model is discussed. Extensions on other datasets, episode-level perturbations, and explicit state-dependent penalty models would test how far these patterns generalize.

5 Conclusion

This paper presented count-based empirical identification of a variable-structure automaton under the P-model on a fixed interaction dataset (3518 steps, 80 episodes). A single reproducible pipeline validated the log, estimated policies and penalties, assessed ε -optimality, checked a simulator, stressed the fit under random row removal, and audited per-state penalties against the global P-model.

The principal outcomes are that α_1 is the lowest-penalty action with weak ε -optimality overall, simulated marginals track the data closely, and global estimates are stable under the chosen subsampling experiment, while per-state penalties show measurable spread that qualifies a strict action-only penalty law. Interpretation, limitations, and extensions are developed in Section 4.4.

APPENDIX

Appendix A: Supplementary source code listing

Excerpts below are taken from the submitted source files, one subsection per module. Line ranges match the repository at report time; together they implement the flow in Section 3.1 (Algorithm 1) and the estimators in Section 2.3.

A.1 fingerprint.py

```
df = load_vsa_dataset(csv_path)

n_rows = len(df)
n_sequences = int(df[CLMN_SEQ_ID].nunique())
lengths = df.groupby(CLMN_SEQ_ID, sort=False).size()
mean_sequence_length = float(lengths.mean())

penalty_rate_per_action = (
    df.groupby(CLMN_ALPHA)[CLMN_BETA].mean().sort_index()
)
penalty_rate_per_action.index.name = CLMN_ALPHA
penalty_rate_per_action.name = "empirical_penalty_rate"

checksum = int(df[CLMN_STEP].sum())
```

A.2 estimation.py

```
# count by grouping how many action alpha taken in state phi
N_phi_alpha = (
    df.groupby([CLMN_PHI_SRC, CLMN_ALPHA]).size()
    .unstack(fill_value=0)
    .reindex(index=states, columns=actions, fill_value=0)
)
N_phi_alpha.index.name = CLMN_PHI_SRC
N_phi_alpha.columns.name = CLMN_ALPHA

# count by grouping how many transitions from state phi_src to state phi_dst for action a
N_phi_alpha_phi: Dict[str, pd.DataFrame] = {}
for a in actions:
    tab = (
        df[df[CLMN_ALPHA] == a]
        .groupby([CLMN_PHI_SRC, CLMN_PHI_DST]).size()
        .unstack(fill_value=0)
        .reindex(index=states, columns=states, fill_value=0)
    )
    tab.index.name = CLMN_PHI_SRC
    tab.columns.name = CLMN_PHI_DST
    N_phi_alpha_phi[a] = tab

# count by grouping how many action alpha valuated with beta
N_alpha_beta = (
    df.groupby([CLMN_ALPHA, CLMN_BETA]).size()
    .unstack(fill_value=0)
    .reindex(index=actions, columns=betas, fill_value=0)
)
N_alpha_beta.index.name = CLMN_ALPHA
N_alpha_beta.columns.name = CLMN_BETA
```

```

# Conditional probabilities -----
# N_phi_alpha : how many action alpha taken in state phi
# N_phi_alpha_phi : how many transitions from state phi_src to state phi_dst for action a
# N_alpha_beta : how many action alpha valuated with beta

row_totals_phi = N_phi_alpha.sum(axis=1)
policy = N_phi_alpha.div(row_totals_phi.replace(0, np.nan), axis=0).fillna(0.0)

transition: Dict[str, pd.DataFrame] = {}
for a in actions:
    N = N_phi_alpha_phi[a]
    T = N.div(N.sum(axis=1).replace(0, np.nan), axis=0).fillna(0.0)
    transition[a] = T

total_per_action = N_alpha_beta.sum(axis=1).replace(0, np.nan)
penalty = (N_alpha_beta[1] / total_per_action).fillna(0.0)
penalty.name = "c_k"

N_phi_alpha_beta = (
    df.groupby([CLMN_PHI_SRC, CLMN_ALPHA, CLMN_BETA]).size()
    .unstack(fill_value=0)
    .reindex(columns=betas, fill_value=0)
)
totals_phi_alpha = N_phi_alpha_beta.sum(axis=1).replace(0, np.nan)
penalty_per_state = (
    (N_phi_alpha_beta[1] / totals_phi_alpha)
    .fillna(0.0)
    .unstack(CLMN_ALPHA)
    .reindex(index=states, columns=actions, fill_value=0.0)
)
penalty_per_state.index.name = CLMN_PHI_SRC
penalty_per_state.columns.name = CLMN_ALPHA

empirical_penalty = (
    df.groupby(CLMN_ALPHA)[CLMN_BETA].mean().reindex(actions, fill_value=0.0)
)
empirical_penalty.name = "empirical_penalty"

# Consistency checks -----
nonzero_phi = row_totals_phi > 0
policy_rows_ok = bool(
    np.allclose(policy.loc[nonzero_phi].sum(axis=1).values, 1.0, atol=1e-9)
)

transition_rows_ok = True
for a, T in transition.items():
    N = N_phi_alpha_phi[a]
    nonzero = N.sum(axis=1) > 0
    if not np.allclose(T.loc[nonzero].sum(axis=1).values, 1.0, atol=1e-9):
        transition_rows_ok = False
        break

penalty_matches_empirical = bool(
    np.allclose(penalty.values, empirical_penalty.values, atol=1e-9)
)

```

A.3 optimal_action_identification.py

```

c_min = float(penalty.min())
ties: List[str] = [str(a) for a in penalty.index[np.isclose(penalty.values, c_min)]]
alpha_star = ties[0] # deterministic pick: first action tied at the minimum

```

```

ranking = (
    penalty.sort_values(ascending=True)
        .to_frame()
        .assign(rank=lambda d: np.arange(1, len(d) + 1, dtype=int))
)
ranking["is_optimal"] = np.isclose(ranking["c_k"].values, c_min)
ranking.index.name = CLMN_ALPHA

result = OptimalActionResult(
    alpha_star=alpha_star,
    c_star=c_min,
    ties=ties,
    ranking=ranking,
    penalty=penalty,
    source=source,
)

```

A.4 e_optimality_analysis.py

```

def _compute_core(
    policy: pd.DataFrame,
    counts_phi_alpha: pd.DataFrame,
    alpha_star: str,
) -> Tuple[float, float, pd.Series, List[str]]:
    if alpha_star not in policy.columns:
        raise ValueError(
            f"alpha* column {alpha_star!r} not in policy columns {list(policy.columns)}. "
            "Regenerate policy.csv / run Task 1 after Task 2."
        )
    row_totals = counts_phi_alpha.sum(axis=1)
    visited = row_totals > 0
    if not bool(visited.any()):
        raise ValueError("No visited states in count matrix; cannot compute epsilon.")

    p_col = policy.loc[visited, alpha_star].astype(float)
    min_p = float(np.min(p_col.values))
    epsilon = 1.0 - min_p
    tol = 1e-9
    argmin_states = [str(s) for s in p_col.index[np.isclose(p_col.values, min_p, atol=tol)]]
    return epsilon, min_p, p_col, argmin_states

```

A.5 simulation.py

```

def _simulate_block(
    real_df: pd.DataFrame,
    policy: pd.DataFrame,
    transition: Dict[str, pd.DataFrame],
    penalty: pd.Series,
    rng: np.random.Generator,
) -> pd.DataFrame:
    """Generate synthetic rows matching real (sequence_id, step) layout."""
    meta = (
        real_df.sort_values([CLMN_SEQ_ID, CLMN_STEP])
        .groupby(CLMN_SEQ_ID, sort=False)
        .agg(length=(CLMN_STEP, "count"), phi0=(CLMN_PHI_SRC, "first"))
    )
    lengths = meta["length"].astype(int).to_numpy()
    phi0s = meta["phi0"].astype(str).to_numpy()
    seq_ids = meta.index.to_numpy()

```

```

states = list(policy.index.astype(str))
actions = list(policy.columns.astype(str))

rows: List[dict] = []
for sid, L, phi in zip(seq_ids, lengths, phi0s):
    phi = str(phi)
    if phi not in policy.index:
        phi = states[0]
    for step in range(1, int(L) + 1):
        prow = policy.loc[phi].to_numpy(dtype=float)
        a = _sample_from_row(rng, actions, prow)
        tmat = transition[a]
        labels = list(tmat.columns.astype(str))
        if phi in tmat.index:
            trow = tmat.loc[phi].to_numpy(dtype=float)
        else:
            trow = np.ones(len(labels), dtype=float) / len(labels)
        phi_next = _sample_from_row(rng, labels, trow)
        p_pen = float(penalty.reindex([a]).fillna(0.0).iloc[0])
        p_pen = min(max(p_pen, 0.0), 1.0)
        beta = int(rng.binomial(1, p_pen))
        rows.append(
            {
                CLMN_SEQ_ID: int(sid),
                CLMN_STEP: step,
                CLMN_PHI_SRC: phi,
                CLMN_PHI_DST: phi_next,
                CLMN_ALPHA: a,
                CLMN_BETA: beta,
            }
        )
        phi = phi_next

return pd.DataFrame(rows)

```

A.6 main.py

`_drop_rows_randomly:`

```

def _drop_rows_randomly(
    df: pd.DataFrame,
    fraction_remove: float,
    rng: np.random.Generator,
) -> Tuple[pd.DataFrame, int]:
    """Return a copy of 'df' with 'round(fraction_remove * n)' rows removed."""
    n = len(df)
    if n <= 1 or fraction_remove <= 0.0:
        return df.copy(), 0
    n_remove = int(round(fraction_remove * n))
    n_remove = min(max(0, n_remove), n - 1)
    if n_remove <= 0:
        return df.copy(), 0
    drop_positions = rng.choice(n, size=n_remove, replace=False)
    out = df.drop(df.index[drop_positions]).reset_index(drop=True)
    return out, n_remove

```

`_run_tasks_1_to_4:`

```

def _run_tasks_1_to_4(
    df: pd.DataFrame,
    csv_path: Path,
    *,

```

```

dataset_label: str,
print_to_console: bool,
) -> PipelineRunResult:
    estimation = run_estimation(
        df,
        dataset_label=dataset_label,
        output_dir=None,
        print_to_console=print_to_console,
        print_penalty_per_state=False,
    )
    optimal = run_optimal_action(
        estimation=estimation,
        output_dir=None,
        print_to_console=print_to_console,
    )
    epsilon_res = run_epsilon_optimality(
        estimation=estimation,
        optimal=optimal,
        output_dir=None,
        print_to_console=print_to_console,
    )
    sim_res = run_simulation(
        estimation=estimation,
        csv_path=csv_path,
        output_dir=None,
        print_to_console=print_to_console,
    )
    return PipelineRunResult(
        label=dataset_label,
        n_rows=len(df),
        alpha_star=optimal.alpha_star,
        c_star=optimal.c_star,
        c_k_per_state=estimation.penalty_per_state.copy(),
        epsilon=epsilon_res.epsilon,
        min_p_alpha_star=epsilon_res.min_p_alpha_star,
        l1_action=sim_res.l1_action,
        l1_phi_src=sim_res.l1_phi_src,
        l1_phi_dst=sim_res.l1_phi_dst,
    )

```

`_print_c_k_per_state_runs:`

```

def _print_c_k_per_state_runs(runs: List[PipelineRunResult]) -> None:
    """Print per-state penalty tables for one or more pipeline runs."""
    lines = [
        _section("Task 6 --Model assumption check"),
        "c_k(phi_i) = P(beta=1 | alpha_k, phi_i)",
        "",
    ]
    for run in runs:
        lines.extend(_format_c_k_per_state_lines(run))
    print("\n".join(lines))

```

`_print_experiments_summary` (spread and table-delta helpers):

```

def _spread_across_states(c_k_per_state: pd.DataFrame) -> pd.Series:
    """Per action: max_phi c_k(phi) - min_phi c_k(phi) (P-model spread diagnostic)."""
    return c_k_per_state.max(axis=0) - c_k_per_state.min(axis=0)

def _max_abs_table_delta(reference: pd.DataFrame, other: pd.DataFrame) -> float:
    """Largest |c_k(phi, alpha) - reference| over aligned cells."""
    aligned = other.reindex_like(reference).fillna(0.0)
    return float(np.abs(aligned - reference).max().max())

```



```

    )

    lines.extend(
        [
            "Spread across states per action (max_phi c_k - min_phi c_k):",
            " Baseline: "
            + " ".join(f"{a}={baseline_spread[a]:.6f}" for a in actions),
        ]
    )

    if perturbed:
        if len(perturbed) > 1:
            spread_mean = perturbed_spreads.mean(axis=0)
            spread_std = perturbed_spreads.std(axis=0, ddof=1)
            lines.append(
                " Perturbed: "
                + " ".join(
                    f"{a}=mean {spread_mean[i]:.6f} std {spread_std[i]:.6f}"
                    for i, a in enumerate(actions)
                )
            )
        lines.extend(
            [
                "",
                "|c_k_per_state - baseline| max cell delta:",
                f" mean={table_deltas.mean():.6f} "
                f"std={table_deltas.std(ddof=1):.6f} "
                f"max={table_deltas.max():.6f}",
            ]
        )
    else:
        p_spread = _spread_across_states(perturbed[0].c_k_per_state)
        lines.append(
            " Perturbed: "
            + " ".join(f"{a}={p_spread[a]:.6f}" for a in actions)
        )
        lines.append(
            f"\n |c_k_per_state - baseline| max cell delta: "
            f"{table_deltas[0]:.6f}"
        )

print("\n".join(lines))
_print_c_k_per_state_runs([baseline, *perturbed])

```

run_all:

```

run_fingerprint(
    csv_path,
    output_dir=None,
    print_to_console=print_to_console,
)

# first run without perturbation
baseline = _run_tasks_1_to_4(
    df_full,
    csv_path,
    dataset_label=f"{csv_path.name} (baseline)",
    print_to_console=print_to_console,
)
results: List[PipelineRunResult] = [baseline]
if print_to_console:
    _print_run_results(baseline)

if perturbation_rate > 0:

```

```

fraction = perturbation_rate / 100.0
rng_master = np.random.default_rng(random_seed)
for t in range(experiment_count):
    trial_seed = int(rng_master.integers(0, 2**31 - 1))
    rng = np.random.default_rng(trial_seed)
    df_p, n_removed = _drop_rows_randomly(df_full, fraction, rng)
    perturbation_run = _run_tasks_1_to_4(
        df_p,
        csv_path,
        dataset_label=f"{csv_path.name} (trial {t}, removed {n_removed})",
        print_to_console=print_to_console,
    )
    if print_to_console:
        _print_run_results(perturbation_run)
    results.append(perturbation_run)

    if print_to_console:
        _print_experiments_summary(
            csv_path=csv_path,
            n_full=len(df_full),
            perturbation_rate=perturbation_rate,
            experiment_count=experiment_count,
            baseline=baseline,
            perturbed=results[1:],
        )
elif print_to_console:
    _print_c_k_per_state_runs([baseline])

```

A.7 util.py

Column constants, shared result types, and CSV loading:

```

# DataFrame / CSV column names (headerless file column order matches COLUMNS)
CLMN_SEQ_ID = "sequence_id"
CLMN_STEP = "step"
CLMN_PHI_SRC = "phi_src"
CLMN_PHI_DST = "phi_dst"
CLMN_ALPHA = "alpha"
CLMN_BETA = "beta"

COLUMNS = [
    CLMN_SEQ_ID,
    CLMN_STEP,
    CLMN_PHI_SRC,
    CLMN_PHI_DST,
    CLMN_ALPHA,
    CLMN_BETA,
]

@dataclass
class EstimationResult:
    """Structured return value of 'estimation.run_estimation'."""

    counts_phi_alpha: pd.DataFrame # N(phi_i, alpha_k)
    counts_phi_alpha_phi: Dict[str, pd.DataFrame] # N(phi_i, alpha_k, phi_j) per alpha
    counts_alpha_beta: pd.DataFrame # N(alpha_k, beta)
    policy: pd.DataFrame # P(alpha_k | phi_i)
    transition: Dict[str, pd.DataFrame] # P(phi_j | phi_i, alpha_k) per alpha
    penalty: pd.Series # c_k = P(beta=1 | alpha_k)
    penalty_per_state: pd.DataFrame # c_k(phi_i) = P(beta=1 | alpha_k, phi_i)
    empirical_penalty: pd.Series # groupby(CLMN_ALPHA)[CLMN_BETA].mean()

```

```

checks: Dict[str, bool] # named pass/fail flags
states: list
actions: list
n_rows: int

```

```

@dataclass
class OptimalActionResult:
    """Structured return value of 'optimal_action_identification.run_optimal_action'."""

```

```

c_star: float # its penalty probability (the minimum c_k)
ties: List[str] # all actions tied at the minimum (usually just [alpha_star])
ranking: pd.DataFrame # actions sorted by c_k (ascending), columns: c_k, rank
penalty: pd.Series # the c_k vector echoed back (action -> c_k)
source: str # "estimation-result" or absolute path to penalty.csv

```

```

def load_vsa_dataset(csv_path: Union[Path, str]) -> pd.DataFrame:
    """Load the headerless UTF-8 VSA interaction CSV.

    Each row is
    '(CLMN_SEQ_ID, CLMN_STEP, CLMN_PHI_SRC, CLMN_PHI_DST, CLMN_ALPHA, CLMN_BETA)'
    with integer 'CLMN_SEQ_ID', 'CLMN_STEP', and 'CLMN_BETA'; state and
    action columns stay as strings (e.g. Greek labels).
    """
    csv_path = Path(csv_path)
    df = pd.read_csv(csv_path, header=None, names=COLUMNS, encoding="utf-8")
    df[CLMN_SEQ_ID] = df[CLMN_SEQ_ID].astype(int)
    df[CLMN_STEP] = df[CLMN_STEP].astype(int)
    df[CLMN_BETA] = df[CLMN_BETA].astype(int)
    return df

```

Appendix B: Full experimental log

Complete console output from python main.py (load_all: perturbation_rate=20, ten trials, random_seed=42). Greek labels are transliterated for typesetting ($a_1 \equiv \alpha_1$, $p_1 \equiv \phi_1$); the raw log is experiments.txt.

```

D:\Documents\00-PhD\04-CSE8003\assignments\term_project\proj_python\.venv\Scripts\python.exe D:\Documents
\00-PhD\04-CSE8003\assignments\term_project\proj_python\main.py

=====
Dataset fingerprint (Engin-vsa_dataset_full.csv)
=====
CSV path : D:\Documents\00-PhD\04-CSE8003\assignments\term_project\Engin-vsa_dataset_full.csv
Total number of rows : 3518
Number of sequences : 80
Mean sequence length : 43.975000

Empirical penalty rate per action (mean of beta given alpha):
alpha
a1 0.199608
a2 0.381579
a3 0.517606

Checksum (sum of step over all rows):
checksum = 83035

=====
Task 1 - Estimation (dataset: Engin-vsa_dataset_full.csv (baseline))
=====
Rows in dataset : 3518
States observed : ['p1', 'p2', 'p3', 'p4', 'p5']
Actions observed: ['a1', 'a2', 'a3']

=====

```

```

N(phi_i, alpha_k) -- raw counts
=====
alpha a1 a2 a3
phi_src
p1 573 78 39
p2 665 157 58
p3 511 157 58
p4 459 154 66
p5 342 138 63

=====
N(phi_i, alpha_k=a1, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 225 228 40 37 43
p2 198 197 192 45 33
p3 36 166 134 140 35
p4 28 40 131 128 132
p5 29 33 22 131 127

=====
N(phi_i, alpha_k=a2, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 27 32 8 9 2
p2 43 54 42 9 9
p3 10 52 45 43 7
p4 14 18 43 32 47
p5 15 13 10 51 49

=====
N(phi_i, alpha_k=a3, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 16 12 3 4 4
p2 20 15 18 2 3
p3 4 16 21 14 3
p4 5 4 20 17 20
p5 8 7 4 24 20

=====
N(alpha_k, beta) -- raw counts
=====
beta 0 1
alpha
a1 2041 509
a2 423 261
a3 137 147

=====
Policy P(alpha_k | phi_i)
=====
alpha a1 a2 a3
phi_src
p1 0.830435 0.113043 0.056522
p2 0.755682 0.178409 0.065909
p3 0.703857 0.216253 0.079890
p4 0.675994 0.226804 0.097202
p5 0.629834 0.254144 0.116022

=====
Transition P(phi_j | phi_i, alpha_k=a1)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.392670 0.397906 0.069808 0.064572 0.075044
p2 0.297744 0.296241 0.288722 0.067669 0.049624
p3 0.070450 0.324853 0.262231 0.273973 0.068493
p4 0.061002 0.087146 0.285403 0.278867 0.287582
p5 0.084795 0.096491 0.064327 0.383041 0.371345

```

```

=====
Transition P(phi_j | phi_i, alpha_k=a2)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.346154 0.410256 0.102564 0.115385 0.025641
p2 0.273885 0.343949 0.267516 0.057325 0.057325
p3 0.063694 0.331210 0.286624 0.273885 0.044586
p4 0.090909 0.116883 0.279221 0.207792 0.305195
p5 0.108696 0.094203 0.072464 0.369565 0.355072

=====
Transition P(phi_j | phi_i, alpha_k=a3)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
118 p1 0.410256 0.307692 0.076923 0.102564 0.102564
p2 0.344828 0.258621 0.310345 0.034483 0.051724
p3 0.068966 0.275862 0.362069 0.241379 0.051724
p4 0.075758 0.060606 0.303030 0.257576 0.303030
p5 0.126984 0.111111 0.063492 0.380952 0.317460

=====
Penalty probabilities c_k = P(beta=1 | alpha_k)
=====
c_k empirical_penalty
alpha
a1 0.199608 0.199608
a2 0.381579 0.381579
a3 0.517606 0.517606

=====
Consistency checks
=====
[PASS] sum_k P(alpha_k | phi_i) == 1
[PASS] sum_j P(phi_j | phi_i, alpha_k) == 1
[PASS] c_k matches empirical frequencies

=====
Task 2 - Optimal Action Identification (source: estimation-result)
=====
Penalty probabilities c_k = P(beta = 1 | alpha_k):
alpha
a1 0.199608
a2 0.381579
a3 0.517606

alpha* = a1
c_* = 0.199608

=====
Ranking by c_k (ascending)
=====
c_k rank is_optimal
alpha
a1 0.199608 1 True
a2 0.381579 2 False
a3 0.517606 3 False

=====
Task 3 - e-optimality (source: estimation-result + optimal-result)
=====
a* = a1
min_i P(a*|p_i) over visited states = 0.629834
e = 1 - min_i P(a*|p_i) = 0.370166
Visited states (count): 5
168 Argmin state(s) (ties): ['p5']

P(a*|p_i) on visited states:
phi_src
p1 0.830435
p2 0.755682

```

```

p3 0.703857
p4 0.675994
p5 0.629834

=====
Interpretation (assignment prompts)
=====
* Is the system strongly optimal?
  Weakly - e is large: at least one visited state rarely chooses a* under the empirical policy.
* Is it noisy?
  Yes - high e and/or high spread across states suggests heterogeneous or stochastic-looking action
  choices.
* Is behavior consistent across states?
  No - behavior differs markedly across visited states (large spread in  $P(a|p)$ ).

Diagnostics: spread(max-min P)=0.200601, std(P)=0.077124

=====
Task 4 - Simulation (source: estimation-result, seed=42)
=====
Real rows: 3518 | Synthetic rows: 3518

|P_data - P_sim| - action distribution
alpha
a1 0.007391
a2 0.007959
a3 0.000569
  L1 sum of abs diffs (actions): 0.015918

|P_data - P_sim| - phi_src visitation
phi_src
p1 0.003127
p2 0.004548
p3 0.001990
p4 0.001137
p5 0.004548
  L1 sum: 0.015350

|P_data - P_sim| - phi_dst visitation
phi_dst
p1 0.004548
p2 0.003695
p3 0.003127
p4 0.000569
p5 0.004548
  L1 sum: 0.016487

|mean_beta_data - mean_beta_sim| per action (penalty rate)
alpha
a1 0.012496
a2 0.005616
a3 0.020856
  L1 sum (per-action |D|): 0.038968

Global penalty rate |D|: data=0.260659 sim=0.252985 |D|=0.007675

=====
main: Engin-vsa_dataset_full.csv (baseline) - Tasks 1--4 finished
=====
  rows processed : 3518
  optimal action : alpha* = a1 (c* = 0.199608)
  epsilon (Task 3) : e = 0.370166 (min  $P(a|p)$  = 0.629834)
  simulation L1 : actions=0.0159 phi_src=0.0153 phi_dst=0.0165

=====
Task 1 - Estimation (dataset: Engin-vsa_dataset_full.csv (trial 0, removed 704))
=====
Rows in dataset : 2814
States observed : ['p1', 'p2', 'p3', 'p4', 'p5']
Actions observed: ['a1', 'a2', 'a3']
=====

```

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```

N(phi_i, alpha_k) -- raw counts
=====
alpha a1 a2 a3
phi_src
p1 453 67 31
p2 529 128 50
p3 401 137 45
p4 360 128 51
p5 264 118 52

=====
N(phi_i, alpha_k=a1, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 171 188 28 32 34
p2 166 155 149 34 25
p3 24 136 109 102 30
p4 22 37 105 98 98
p5 25 26 16 100 97

=====
268 N(phi_i, alpha_k=a2, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 25 27 6 7 2
p2 35 49 32 8 4
p3 9 48 40 34 6
p4 11 14 35 29 39
p5 10 10 8 45 45

=====
N(phi_i, alpha_k=a3, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 14 9 2 3 3
p2 16 12 17 2 3
p3 1 15 15 12 2
p4 5 3 16 14 13
p5 7 6 4 19 16

=====
N(alpha_k, beta) -- raw counts
=====
beta 0 1
alpha
a1 1601 406
a2 353 225
a3 110 119

=====
Policy P(alpha_k | phi_i)
=====
alpha a1 a2 a3
phi_src
p1 0.822142 0.121597 0.056261
p2 0.748232 0.181047 0.070721
p3 0.687822 0.234991 0.077187
p4 0.667904 0.237477 0.094620
p5 0.608295 0.271889 0.119816

=====
Transition P(phi_j | phi_i, alpha_k=a1)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.377483 0.415011 0.061810 0.070640 0.075055
p2 0.313800 0.293006 0.281664 0.064272 0.047259
p3 0.059850 0.339152 0.271820 0.254364 0.074813
p4 0.061111 0.102778 0.291667 0.272222 0.272222
318 p5 0.094697 0.098485 0.060606 0.378788 0.367424

```

```

=====
Transition P(phi_j | phi_i, alpha_k=a2)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.373134 0.402985 0.089552 0.104478 0.029851
p2 0.273438 0.382812 0.250000 0.062500 0.031250
p3 0.065693 0.350365 0.291971 0.248175 0.043796
p4 0.085938 0.109375 0.273438 0.226562 0.304688
p5 0.084746 0.084746 0.067797 0.381356 0.381356

=====
Transition P(phi_j | phi_i, alpha_k=a3)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.451613 0.290323 0.064516 0.096774 0.096774
p2 0.320000 0.240000 0.340000 0.040000 0.060000
p3 0.022222 0.333333 0.333333 0.266667 0.044444
p4 0.098039 0.058824 0.313725 0.274510 0.254902
p5 0.134615 0.115385 0.076923 0.365385 0.307692

=====
Penalty probabilities c_k = P(beta=1 | alpha_k)
=====
c_k empirical_penalty
alpha
a1 0.202292 0.202292
a2 0.389273 0.389273
a3 0.519651 0.519651

=====
Consistency checks
=====
[PASS] sum_k P(alpha_k | phi_i) == 1
[PASS] sum_j P(phi_j | phi_i, alpha_k) == 1
[PASS] c_k matches empirical frequencies

=====
Task 2 - Optimal Action Identification (source: estimation-result)
=====
Penalty probabilities c_k = P(beta = 1 | alpha_k):
alpha
a1 0.202292
a2 0.389273
a3 0.519651

alpha* = a1
c_* = 0.202292

=====
Ranking by c_k (ascending)
=====
c_k rank is_optimal
alpha
a1 0.202292 1 True
a2 0.389273 2 False
a3 0.519651 3 False

=====
Task 3 - e-optimality (source: estimation-result + optimal-result)
=====
a* = a1
min_i P(a*|p_i) over visited states = 0.608295
e = 1 - min_i P(a*|p_i) = 0.391705
Visited states (count): 5
Argmin state(s) (ties): ['p5']

P(a*|p_i) on visited states:
phi_src
p1 0.822142
p2 0.748232

```

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```

p3 0.687822
p4 0.667904
p5 0.608295

=====
Interpretation (assignment prompts)
=====
* Is the system strongly optimal?
  Weakly - e is large: at least one visited state rarely chooses a* under the empirical policy.
* Is it noisy?
  Yes - high e and/or high spread across states suggests heterogeneous or stochastic-looking action
  choices.
* Is behavior consistent across states?
  No - behavior differs markedly across visited states (large spread in P(a*|p)).

Diagnostics: spread(max-min P)=0.213847, std(P)=0.081542

=====
Task 4 - Simulation (source: estimation-result, seed=42)
=====
Real rows: 3518 | Synthetic rows: 3518

|P_data - P_sim| - action distribution
alpha
a1 0.002843
a2 0.001137
a3 0.001706
  L1 sum of abs diffs (actions): 0.005685

|P_data - P_sim| - phi_src visitation
phi_src
p1 0.004832
p2 0.020182
p3 0.002843
p4 0.009949
p5 0.012223
  L1 sum: 0.050028

|P_data - P_sim| - phi_dst visitation
phi_dst
p1 0.006822
p2 0.018761
p3 0.003127
p4 0.010233
p5 0.012223
  L1 sum: 0.051165

|mean_beta_data - mean_beta_sim| per action (penalty rate)
alpha
a1 0.005513
a2 0.006502
a3 0.020325
  L1 sum (per-action |D|): 0.032341

Global penalty rate |D|: data=0.260659 sim=0.260375 |D|=0.000284

=====
main: Engin-vsa_dataset_full.csv (trial 0, removed 704) - Tasks 1--4 finished
=====
  rows processed : 2814
  optimal action : alpha* = a1 (c* = 0.202292)
  epsilon (Task 3) : e = 0.391705 (min P(a*|p) = 0.608295)
  simulation L1 : actions=0.0057 phi_src=0.0500 phi_dst=0.0512

=====
Task 1 - Estimation (dataset: Engin-vsa_dataset_full.csv (trial 1, removed 704))
=====
Rows in dataset : 2814
States observed : ['p1', 'p2', 'p3', 'p4', 'p5']
Actions observed: ['a1', 'a2', 'a3']
=====

```

```

N(phi_i, alpha_k) -- raw counts
=====
alpha a1 a2 a3
phi_src
468 p1 448 62 29
p2 532 134 41
p3 415 126 45
p4 377 122 61
p5 264 108 50

=====
N(phi_i, alpha_k=a1, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 178 180 27 29 34
p2 159 162 150 34 27
p3 30 131 112 114 28
p4 21 36 110 105 105
p5 25 20 16 105 98

=====
N(phi_i, alpha_k=a2, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 20 23 8 9 2
p2 37 44 38 9 6
p3 9 43 35 34 5
p4 12 16 33 24 37
p5 12 12 6 40 38

=====
N(phi_i, alpha_k=a3, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 15 8 3 1 2
p2 12 10 16 1 2
p3 4 13 15 10 3
p4 5 4 19 13 20
p5 8 6 4 15 17

=====
N(alpha_k, beta) -- raw counts
=====
beta 0 1
alpha
a1 1631 405
a2 339 213
a3 107 119

=====
518 Policy P(alpha_k | phi_i)
=====
alpha a1 a2 a3
phi_src
p1 0.831169 0.115028 0.053803
p2 0.752475 0.189533 0.057992
p3 0.708191 0.215017 0.076792
p4 0.673214 0.217857 0.108929
p5 0.625592 0.255924 0.118483

=====
Transition P(phi_j | phi_i, alpha_k=a1)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.397321 0.401786 0.060268 0.064732 0.075893
p2 0.298872 0.304511 0.281955 0.063910 0.050752
p3 0.072289 0.315663 0.269880 0.274699 0.067470
p4 0.055703 0.095491 0.291777 0.278515 0.278515
p5 0.094697 0.075758 0.060606 0.397727 0.371212

```

```

=====
Transition P(phi_j | phi_i, alpha_k=a2)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.322581 0.370968 0.129032 0.145161 0.032258
p2 0.276119 0.328358 0.283582 0.067164 0.044776
p3 0.071429 0.341270 0.277778 0.269841 0.039683
p4 0.098361 0.131148 0.270492 0.196721 0.303279
p5 0.111111 0.111111 0.055556 0.370370 0.351852

=====
Transition P(phi_j | phi_i, alpha_k=a3)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.517241 0.275862 0.103448 0.034483 0.068966
p2 0.292683 0.243902 0.390244 0.024390 0.048780
p3 0.088889 0.288889 0.333333 0.222222 0.066667
p4 0.081967 0.065574 0.311475 0.213115 0.327869
p5 0.160000 0.120000 0.080000 0.300000 0.340000

=====
Penalty probabilities c_k = P(beta=1 | alpha_k)
=====
c_k empirical_penalty
alpha
a1 0.198919 0.198919
a2 0.385870 0.385870
a3 0.526549 0.526549

=====
Consistency checks
=====
[PASS] sum_k P(alpha_k | phi_i) == 1
[PASS] sum_j P(phi_j | phi_i, alpha_k) == 1
[PASS] c_k matches empirical frequencies

=====
Task 2 - Optimal Action Identification (source: estimation-result)
=====
Penalty probabilities c_k = P(beta = 1 | alpha_k):
alpha
a1 0.198919
a2 0.385870
a3 0.526549

alpha* = a1
c_* = 0.198919

=====
Ranking by c_k (ascending)
=====
c_k rank is_optimal
alpha
a1 0.198919 1 True
a2 0.385870 2 False
a3 0.526549 3 False

=====
Task 3 - e-optimality (source: estimation-result + optimal-result)
=====
a* = a1
min_i P(a*|p_i) over visited states = 0.625592
e = 1 - min_i P(a*|p_i) = 0.374408
Visited states (count): 5
Argmin state(s) (ties): ['p5']

P(a*|p_i) on visited states:
phi_src
p1 0.831169
p2 0.752475

```

p3 0.708191
p4 0.673214
p5 0.625592

=====
Interpretation (assignment prompts)
=====

* Is the system strongly optimal?
Weakly - e is large: at least one visited state rarely chooses a* under the empirical policy.
* Is it noisy?
Yes - high e and/or high spread across states suggests heterogeneous or stochastic-looking action choices.
* Is behavior consistent across states?
No - behavior differs markedly across visited states (large spread in $P(a|p)$).

Diagnostics: spread(max-min P)=0.205576, std(P)=0.078481

=====
Task 4 - Simulation (source: estimation-result, seed=42)
=====

Real rows: 3518 | Synthetic rows: 3518

|P_data - P_sim| - action distribution
alpha
a1 0.008812
a2 0.008528
a3 0.000284
L1 sum of abs diffs (actions): 0.017624

|P_data - P_sim| - phi_src visitation
phi_src
p1 0.008812
p2 0.006538
p3 0.004832
p4 0.002274
p5 0.008243
L1 sum: 0.030699

|P_data - P_sim| - phi_dst visitation
phi_dst
p1 0.009949
p2 0.006538
p3 0.005401
p4 0.002843
p5 0.008243
L1 sum: 0.032973

|mean_beta_data - mean_beta_sim| per action (penalty rate)
alpha
a1 0.011696
a2 0.006800
a3 0.026564
L1 sum (per-action |D|): 0.045060

Global penalty rate |D|: data=0.260659 sim=0.253837 |D|=0.006822

=====
main: Engin-vsa_dataset_full.csv (trial 1, removed 704) - Tasks 1--4 finished
=====

rows processed : 2814
optimal action : alpha* = a1 (c* = 0.198919)
epsilon (Task 3) : e = 0.374408 (min $P(a|p)$ = 0.625592)
simulation L1 : actions=0.0176 phi_src=0.0307 phi_dst=0.0330

=====
Task 1 - Estimation (dataset: Engin-vsa_dataset_full.csv (trial 2, removed 704))
=====

Rows in dataset : 2814
States observed : ['p1', 'p2', 'p3', 'p4', 'p5']
Actions observed: ['a1', 'a2', 'a3']
=====

```
N(phi_i, alpha_k) -- raw counts
```

```
=====
alpha a1 a2 a3
phi_src
p1 467 63 31
p2 526 125 54
p3 403 133 47
p4 355 131 46
p5 271 109 53
```

```
=====
N(phi_i, alpha_k=a1, phi_j) -- raw counts
=====
```

```
phi_dst p1 p2 p3 p4 p5
phi_src
p1 182 186 32 32 35
p2 153 152 154 39 28
p3 29 129 102 114 29
p4 16 33 103 100 103
p5 20 27 18 103 103
```

```
=====
N(phi_i, alpha_k=a2, phi_j) -- raw counts
=====
```

```
phi_dst p1 p2 p3 p4 p5
phi_src
p1 25 25 5 6 2
p2 34 41 35 6 9
p3 6 45 39 37 6
p4 14 16 38 24 39
p5 12 11 9 35 42
```

```
=====
N(phi_i, alpha_k=a3, phi_j) -- raw counts
=====
```

```
phi_dst p1 p2 p3 p4 p5
phi_src
p1 15 7 2 4 3
p2 20 14 16 2 2
p3 4 14 16 11 2
p4 4 2 16 10 14
p5 6 6 3 23 15
```

```
=====
N(alpha_k, beta) -- raw counts
=====
```

```
beta 0 1
alpha
a1 1612 410
a2 349 212
a3 115 116
```

```
=====
Policy P(alpha_k | phi_i)
=====
```

```
alpha a1 a2 a3
phi_src
p1 0.832442 0.112299 0.055258
p2 0.746099 0.177305 0.076596
p3 0.691252 0.228130 0.080617
p4 0.667293 0.246241 0.086466
p5 0.625866 0.251732 0.122402
```

```
=====
Transition P(phi_j | phi_i, alpha_k=a1)
=====
```

```
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.389722 0.398287 0.068522 0.068522 0.074946
p2 0.290875 0.288973 0.292776 0.074144 0.053232
p3 0.071960 0.320099 0.253102 0.282878 0.071960
p4 0.045070 0.092958 0.290141 0.281690 0.290141
p5 0.073801 0.099631 0.066421 0.380074 0.380074
```

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```

=====
Transition P(phi_j | phi_i, alpha_k=a2)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.396825 0.396825 0.079365 0.095238 0.031746
p2 0.272000 0.328000 0.280000 0.048000 0.072000
p3 0.045113 0.338346 0.293233 0.278195 0.045113
p4 0.106870 0.122137 0.290076 0.183206 0.297710
p5 0.110092 0.100917 0.082569 0.321101 0.385321

=====
768 Transition P(phi_j | phi_i, alpha_k=a3)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.483871 0.225806 0.064516 0.129032 0.096774
p2 0.370370 0.259259 0.296296 0.037037 0.037037
p3 0.085106 0.297872 0.340426 0.234043 0.042553
p4 0.086957 0.043478 0.347826 0.217391 0.304348
p5 0.113208 0.113208 0.056604 0.433962 0.283019

=====
Penalty probabilities c_k = P(beta=1 | alpha_k)
=====
c_k empirical_penalty
alpha
a1 0.202770 0.202770
a2 0.377897 0.377897
a3 0.502165 0.502165

=====
Consistency checks
=====
[PASS] sum_k P(alpha_k | phi_i) == 1
[PASS] sum_j P(phi_j | phi_i, alpha_k) == 1
[PASS] c_k matches empirical frequencies

=====
Task 2 - Optimal Action Identification (source: estimation-result)
=====
Penalty probabilities c_k = P(beta = 1 | alpha_k):
alpha
a1 0.202770
a2 0.377897
a3 0.502165

alpha* = a1
c_* = 0.202770

=====
Ranking by c_k (ascending)
=====
c_k rank is_optimal
alpha
a1 0.202770 1 True
a2 0.377897 2 False
a3 0.502165 3 False

=====
Task 3 - e-optimality (source: estimation-result + optimal-result)
=====
818 a* = a1
min_i P(a*|p_i) over visited states = 0.625866
e = 1 - min_i P(a*|p_i) = 0.374134
Visited states (count): 5
Argmin state(s) (ties): ['p5']

P(a*|p_i) on visited states:
phi_src
p1 0.832442
p2 0.746099

```

```

p3 0.691252
p4 0.667293
p5 0.625866

=====
Interpretation (assignment prompts)
=====
* Is the system strongly optimal?
  Weakly - e is large: at least one visited state rarely chooses a* under the empirical policy.
* Is it noisy?
  Yes - high e and/or high spread across states suggests heterogeneous or stochastic-looking action
  choices.
* Is behavior consistent across states?
  No - behavior differs markedly across visited states (large spread in  $P(a|p)$ ).

Diagnostics: spread(max-min P)=0.206576, std(P)=0.079868

=====
Task 4 - Simulation (source: estimation-result, seed=42)
=====
Real rows: 3518 | Synthetic rows: 3518

|P_data - P_sim| - action distribution
alpha
a1 0.001421
a2 0.000569
a3 0.000853
  L1 sum of abs diffs (actions): 0.002843

|P_data - P_sim| - phi_src visitation
phi_src
p1 0.003127
p2 0.001990
p3 0.000284
p4 0.001990
p5 0.001137
  L1 sum: 0.008528

|P_data - P_sim| - phi_dst visitation
phi_dst
p1 0.004264
p2 0.002843
p3 0.000284
p4 0.002558
p5 0.001421
  L1 sum: 0.011370

|mean_beta_data - mean_beta_sim| per action (penalty rate)
alpha
a1 0.006681
a2 0.004718
a3 0.018980
  L1 sum (per-action |D|): 0.030379

Global penalty rate |D|: data=0.260659 sim=0.258670 |D|=0.001990

=====
main: Engin-vsa_dataset_full.csv (trial 2, removed 704) - Tasks 1--4 finished
=====
  rows processed : 2814
  optimal action : alpha* = a1 (c* = 0.202770)
  epsilon (Task 3) : e = 0.374134 (min  $P(a|p)$  = 0.625866)
  simulation L1 : actions=0.0028 phi_src=0.0085 phi_dst=0.0114

=====
Task 1 - Estimation (dataset: Engin-vsa_dataset_full.csv (trial 3, removed 704))
=====
Rows in dataset : 2814
States observed : ['p1', 'p2', 'p3', 'p4', 'p5']
Actions observed: ['a1', 'a2', 'a3']
=====

```

```

N(phi_i, alpha_k) -- raw counts
=====
alpha a1 a2 a3
phi_src
p1 461 68 34
p2 529 131 45
p3 411 125 37
p4 360 123 57
p5 281 103 49

=====
N(phi_i, alpha_k=a1, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 179 184 31 32 35
p2 154 153 156 36 30
p3 28 129 112 114 28
918 p4 20 28 102 107 103
p5 24 26 16 110 105

=====
N(phi_i, alpha_k=a2, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 24 29 5 8 2
p2 36 44 35 8 8
p3 9 44 36 32 4
p4 12 16 35 23 37
p5 12 9 8 39 35

=====
N(phi_i, alpha_k=a3, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 13 11 2 4 4
p2 16 13 12 1 3
p3 0 10 17 8 2
p4 5 3 19 13 17
p5 7 4 4 17 17

=====
N(alpha_k, beta) -- raw counts
=====
beta 0 1
alpha
a1 1625 417
a2 339 211
a3 110 112

=====
Policy P(alpha_k | phi_i)
=====
alpha a1 a2 a3
phi_src
p1 0.818828 0.120782 0.060391
p2 0.750355 0.185816 0.063830
p3 0.717277 0.218150 0.064572
p4 0.666667 0.227778 0.105556
p5 0.648961 0.237875 0.113164

=====
Transition P(phi_j | phi_i, alpha_k=a1)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
968 p1 0.388286 0.399132 0.067245 0.069414 0.075922
p2 0.291115 0.289225 0.294896 0.068053 0.056711
p3 0.068127 0.313869 0.272506 0.277372 0.068127
p4 0.055556 0.077778 0.283333 0.297222 0.286111
p5 0.085409 0.092527 0.056940 0.391459 0.373665

```



```

=====
Transition P(phi_j | phi_i, alpha_k=a2)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.352941 0.426471 0.073529 0.117647 0.029412
p2 0.274809 0.335878 0.267176 0.061069 0.061069
p3 0.072000 0.352000 0.288000 0.256000 0.032000
p4 0.097561 0.130081 0.284553 0.186992 0.300813
p5 0.116505 0.087379 0.077670 0.378641 0.339806

=====
Transition P(phi_j | phi_i, alpha_k=a3)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.382353 0.323529 0.058824 0.117647 0.117647
p2 0.355556 0.288889 0.266667 0.022222 0.066667
p3 0.000000 0.270270 0.459459 0.216216 0.054054
p4 0.087719 0.052632 0.333333 0.228070 0.298246
p5 0.142857 0.081633 0.081633 0.346939 0.346939

=====
Penalty probabilities c_k = P(beta=1 | alpha_k)
=====
c_k empirical_penalty
alpha
a1 0.204212 0.204212
a2 0.383636 0.383636
a3 0.504505 0.504505

=====
Consistency checks
=====
[PASS] sum_k P(alpha_k | phi_i) == 1
[PASS] sum_j P(phi_j | phi_i, alpha_k) == 1
[PASS] c_k matches empirical frequencies

=====
Task 2 - Optimal Action Identification (source: estimation-result)
=====
Penalty probabilities c_k = P(beta = 1 | alpha_k):
alpha
a1 0.204212
a2 0.383636
a3 0.504505

alpha* = a1
c_* = 0.204212

=====
Ranking by c_k (ascending)
=====
c_k rank is_optimal
alpha
a1 0.204212 1 True
a2 0.383636 2 False
a3 0.504505 3 False

=====
Task 3 - e-optimality (source: estimation-result + optimal-result)
=====
a* = a1
min_i P(a*|p_i) over visited states = 0.648961
e = 1 - min_i P(a*|p_i) = 0.351039
Visited states (count): 5
Argmin state(s) (ties): ['p5']

P(a*|p_i) on visited states:
phi_src
p1 0.818828
p2 0.750355

```

1018

```

p3 0.717277
p4 0.666667
p5 0.648961

=====
Interpretation (assignment prompts)
=====
* Is the system strongly optimal?
  Weakly - e is large: at least one visited state rarely chooses a* under the empirical policy.
* Is it noisy?
  Yes - high e and/or high spread across states suggests heterogeneous or stochastic-looking action
  choices.
* Is behavior consistent across states?
  Partially - some variation across states but not extreme.

Diagnostics: spread(max-min P)=0.169867, std(P)=0.068165

=====
Task 4 - Simulation (source: estimation-result, seed=42)
=====
Real rows: 3518 | Synthetic rows: 3518

1068 |P_data - P_sim| - action distribution
alpha
a1 0.006822
a2 0.007106
a3 0.000284
  L1 sum of abs diffs (actions): 0.014213

|P_data - P_sim| - phi_src visitation
phi_src
p1 0.000569
p2 0.002274
p3 0.003695
p4 0.000569
p5 0.002558
  L1 sum: 0.009665

|P_data - P_sim| - phi_dst visitation
phi_dst
p1 0.001706
p2 0.003411
p3 0.003695
p4 0.000000
p5 0.001990
  L1 sum: 0.010802

|mean_beta_data - mean_beta_sim| per action (penalty rate)
alpha
a1 0.006912
a2 0.006888
a3 0.005201
  L1 sum (per-action |D|): 0.019001

Global penalty rate |D|: data=0.260659 sim=0.256111 |D|=0.004548

=====
main: Engin-vsa_dataset_full.csv (trial 3, removed 704) - Tasks 1--4 finished
=====
  rows processed : 2814
  optimal action : alpha* = a1 (c* = 0.204212)
  epsilon (Task 3) : e = 0.351039 (min P(a*|p) = 0.648961)
  simulation L1 : actions=0.0142 phi_src=0.0097 phi_dst=0.0108

=====
Task 1 - Estimation (dataset: Engin-vsa_dataset_full.csv (trial 4, removed 704))
=====
Rows in dataset : 2814
States observed : ['p1', 'p2', 'p3', 'p4', 'p5']
Actions observed: ['a1', 'a2', 'a3']
=====

```

```

1118 N(phi_i, alpha_k) -- raw counts
=====
alpha a1 a2 a3
phi_src
p1 448 65 36
p2 530 128 46
p3 405 122 44
p4 378 130 48
p5 273 108 53

=====
N(phi_i, alpha_k=a1, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 179 174 32 31 32
p2 164 160 146 34 26
p3 29 136 108 102 30
p4 21 35 108 106 108
p5 24 25 15 108 101

=====
N(phi_i, alpha_k=a2, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 25 25 6 7 2
p2 37 42 34 8 7
p3 9 37 34 35 7
p4 13 14 37 26 40
p5 13 10 8 40 37

=====
N(phi_i, alpha_k=a3, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 16 11 2 4 3
p2 17 12 13 1 3
p3 3 11 16 11 3
p4 3 3 15 14 13
p5 6 7 2 20 18

=====
N(alpha_k, beta) -- raw counts
=====
beta 0 1
alpha
a1 1632 402
a2 339 214
1168 a3 105 122

=====
Policy P(alpha_k | phi_i)
=====
alpha a1 a2 a3
phi_src
p1 0.816029 0.118397 0.065574
p2 0.752841 0.181818 0.065341
p3 0.709282 0.213660 0.077058
p4 0.679856 0.233813 0.086331
p5 0.629032 0.248848 0.122120

=====
Transition P(phi_j | phi_i, alpha_k=a1)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.399554 0.388393 0.071429 0.069196 0.071429
p2 0.309434 0.301887 0.275472 0.064151 0.049057
p3 0.071605 0.335802 0.266667 0.251852 0.074074
p4 0.055556 0.092593 0.285714 0.280423 0.285714
p5 0.087912 0.091575 0.054945 0.395604 0.369963

```

1218

```

=====
Transition P(phi_j | phi_i, alpha_k=a2)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.384615 0.384615 0.092308 0.107692 0.030769
p2 0.289062 0.328125 0.265625 0.062500 0.054688
p3 0.073770 0.303279 0.278689 0.286885 0.057377
p4 0.100000 0.107692 0.284615 0.200000 0.307692
p5 0.120370 0.092593 0.074074 0.370370 0.342593

=====
Transition P(phi_j | phi_i, alpha_k=a3)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.444444 0.305556 0.055556 0.111111 0.083333
p2 0.369565 0.260870 0.282609 0.021739 0.065217
p3 0.068182 0.250000 0.363636 0.250000 0.068182
p4 0.062500 0.062500 0.312500 0.291667 0.270833
p5 0.113208 0.132075 0.037736 0.377358 0.339623

=====
Penalty probabilities c_k = P(beta=1 | alpha_k)
=====
c_k empirical_penalty
alpha
a1 0.197640 0.197640
a2 0.386980 0.386980
a3 0.537445 0.537445

=====
Consistency checks
=====
[PASS] sum_k P(alpha_k | phi_i) == 1
[PASS] sum_j P(phi_j | phi_i, alpha_k) == 1
[PASS] c_k matches empirical frequencies

=====
Task 2 - Optimal Action Identification (source: estimation-result)
=====
Penalty probabilities c_k = P(beta = 1 | alpha_k):
alpha
a1 0.197640
a2 0.386980
a3 0.537445

alpha* = a1
c_* = 0.197640

=====
Ranking by c_k (ascending)
=====
c_k rank is_optimal
alpha
a1 0.197640 1 True
a2 0.386980 2 False
a3 0.537445 3 False

=====
Task 3 - e-optimality (source: estimation-result + optimal-result)
=====
a* = a1
min_i P(a*|p_i) over visited states = 0.629032
e = 1 - min_i P(a*|p_i) = 0.370968
Visited states (count): 5
Argmin state(s) (ties): ['p5']

P(a*|p_i) on visited states:
phi_src
p1 0.816029
p2 0.752841

```

p3 0.709282
p4 0.679856
p5 0.629032

1268

```
=====
Interpretation (assignment prompts)
=====
* Is the system strongly optimal?
  Weakly - e is large: at least one visited state rarely chooses a* under the empirical policy.
* Is it noisy?
  Yes - high e and/or high spread across states suggests heterogeneous or stochastic-looking action
  choices.
* Is behavior consistent across states?
  Partially - some variation across states but not extreme.
```

Diagnostics: spread(max-min P)=0.186997, std(P)=0.071183

```
=====
Task 4 - Simulation (source: estimation-result, seed=42)
=====
```

Real rows: 3518 | Synthetic rows: 3518

```
|P_data - P_sim| - action distribution
alpha
a1 0.006538
a2 0.008243
a3 0.001706
  L1 sum of abs diffs (actions): 0.016487
```

```
|P_data - P_sim| - phi_src visitation
phi_src
p1 0.011939
p2 0.001706
p3 0.005685
p4 0.004548
p5 0.003411
  L1 sum: 0.027288
```

```
|P_data - P_sim| - phi_dst visitation
phi_dst
p1 0.013360
p2 0.001137
p3 0.007106
p4 0.003980
p5 0.003411
  L1 sum: 0.028994
```

```
|mean_beta_data - mean_beta_sim| per action (penalty rate)
alpha
a1 0.014610
a2 0.012314
a3 0.047912
  L1 sum (per-action |D|): 0.074836
```

1318 Global penalty rate |D|: data=0.260659 sim=0.255259 |D|=0.005401

```
=====
main: Engin-vsa_dataset_full.csv (trial 4, removed 704) - Tasks 1--4 finished
=====
```

```
rows processed : 2814
optimal action : alpha* = a1 (c* = 0.197640)
epsilon (Task 3) : e = 0.370968 (min P(a*|p) = 0.629032)
simulation L1 : actions=0.0165 phi_src=0.0273 phi_dst=0.0290
```

```
=====
Task 1 - Estimation (dataset: Engin-vsa_dataset_full.csv (trial 5, removed 704))
=====
```

```
Rows in dataset : 2814
States observed : ['p1', 'p2', 'p3', 'p4', 'p5']
Actions observed: ['a1', 'a2', 'a3']
=====
```

1368

```

N(phi_i, alpha_k) -- raw counts
=====
alpha a1 a2 a3
phi_src
p1 466 60 27
p2 548 127 49
p3 419 124 44
p4 347 122 55
p5 258 116 52

=====
N(phi_i, alpha_k=a1, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 179 191 30 31 35
p2 162 160 160 37 29
p3 30 136 110 112 31
p4 22 25 108 87 105
p5 19 25 16 104 94

=====
N(phi_i, alpha_k=a2, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 22 25 7 5 1
p2 36 40 37 8 6
p3 8 40 33 37 6
p4 13 15 33 26 35
p5 13 10 10 45 38

=====
N(phi_i, alpha_k=a3, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 9 9 3 3 3
p2 17 13 17 1 1
p3 2 12 14 13 3
p4 3 3 17 15 17
p5 7 6 4 19 16

=====
N(alpha_k, beta) -- raw counts
=====
beta 0 1
alpha
a1 1635 403
a2 346 203
a3 108 119

=====
Policy P(alpha_k | phi_i)
=====
alpha a1 a2 a3
phi_src
p1 0.842676 0.108499 0.048825
p2 0.756906 0.175414 0.067680
p3 0.713799 0.211244 0.074957
p4 0.662214 0.232824 0.104962
p5 0.605634 0.272300 0.122066

=====
Transition P(phi_j | phi_i, alpha_k=a1)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.384120 0.409871 0.064378 0.066524 0.075107
p2 0.295620 0.291971 0.291971 0.067518 0.052920
p3 0.071599 0.324582 0.262530 0.267303 0.073986
p4 0.063401 0.072046 0.311239 0.250720 0.302594
p5 0.073643 0.096899 0.062016 0.403101 0.364341

```

1418

```

=====
Transition P(phi_j | phi_i, alpha_k=a2)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.366667 0.416667 0.116667 0.083333 0.016667
p2 0.283465 0.314961 0.291339 0.062992 0.047244
p3 0.064516 0.322581 0.266129 0.298387 0.048387
p4 0.106557 0.122951 0.270492 0.213115 0.286885
p5 0.112069 0.086207 0.086207 0.387931 0.327586

=====
Transition P(phi_j | phi_i, alpha_k=a3)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.333333 0.333333 0.111111 0.111111 0.111111
p2 0.346939 0.265306 0.346939 0.020408 0.020408
p3 0.045455 0.272727 0.318182 0.295455 0.068182
p4 0.054545 0.054545 0.309091 0.272727 0.309091
p5 0.134615 0.115385 0.076923 0.365385 0.307692

=====
Penalty probabilities c_k = P(beta=1 | alpha_k)
=====
c_k empirical_penalty
alpha
a1 0.197743 0.197743
a2 0.369763 0.369763
a3 0.524229 0.524229

=====
Consistency checks
=====
[PASS] sum_k P(alpha_k | phi_i) == 1
[PASS] sum_j P(phi_j | phi_i, alpha_k) == 1
[PASS] c_k matches empirical frequencies

=====
Task 2 - Optimal Action Identification (source: estimation-result)
=====
Penalty probabilities c_k = P(beta = 1 | alpha_k):
alpha
a1 0.197743
a2 0.369763
a3 0.524229

alpha* = a1
c_* = 0.197743

=====
Ranking by c_k (ascending)
=====
c_k rank is_optimal
alpha
a1 0.197743 1 True
a2 0.369763 2 False
a3 0.524229 3 False

=====
Task 3 - e-optimality (source: estimation-result + optimal-result)
=====
a* = a1
min_i P(a*|p_i) over visited states = 0.605634
e = 1 - min_i P(a*|p_i) = 0.394366
Visited states (count): 5
Argmin state(s) (ties): ['p5']

P(a*|p_i) on visited states:
phi_src
p1 0.842676
p2 0.756906

```

1468

```

p3 0.713799
p4 0.662214
p5 0.605634

=====
Interpretation (assignment prompts)
=====
* Is the system strongly optimal?
  Weakly - e is large: at least one visited state rarely chooses a* under the empirical policy.
* Is it noisy?
  Yes - high e and/or high spread across states suggests heterogeneous or stochastic-looking action
  choices.
* Is behavior consistent across states?
  No - behavior differs markedly across visited states (large spread in P(a*|p)).

Diagnostics: spread(max-min P)=0.237043, std(P)=0.090552

=====
Task 4 - Simulation (source: estimation-result, seed=42)
=====
Real rows: 3518 | Synthetic rows: 3518

|P_data - P_sim| - action distribution
alpha
a1 0.004264
a2 0.003980
a3 0.000284
  L1 sum of abs diffs (actions): 0.008528

|P_data - P_sim| - phi_src visitation
phi_src
p1 0.000569
p2 0.002558
p3 0.007391
p4 0.003411
p5 0.001990
  L1 sum: 0.015918

|P_data - P_sim| - phi_dst visitation
phi_dst
p1 0.001990
p2 0.003411
p3 0.007106
p4 0.003980
p5 0.001706
  L1 sum: 0.018192

|mean_beta_data - mean_beta_sim| per action (penalty rate)
alpha
a1 0.015202
a2 0.011430
a3 0.026564
  L1 sum (per-action |D|): 0.053196

Global penalty rate |D|: data=0.260659 sim=0.248721 |D|=0.011939

=====
main: Engin-vsa_dataset_full.csv (trial 5, removed 704) - Tasks 1--4 finished
=====
  rows processed : 2814
  optimal action : alpha* = a1 (c* = 0.197743)
  epsilon (Task 3) : e = 0.394366 (min P(a*|p) = 0.605634)
  simulation L1 : actions=0.0085 phi_src=0.0159 phi_dst=0.0182

=====
Task 1 - Estimation (dataset: Engin-vsa_dataset_full.csv (trial 6, removed 704))
=====
Rows in dataset : 2814
States observed : ['p1', 'p2', 'p3', 'p4', 'p5']
Actions observed: ['a1', 'a2', 'a3']
=====

```

1518


```

N(phi_i, alpha_k) -- raw counts
=====
alpha a1 a2 a3
phi_src
p1 452 64 32
p2 539 123 46
p3 395 123 46
p4 371 124 52
p5 288 112 47

=====
N(phi_i, alpha_k=a1, phi_j) -- raw counts
=====
1568 phi_dst p1 p2 p3 p4 p5
phi_src
p1 181 174 32 31 34
p2 160 160 156 37 26
p3 26 131 102 111 25
p4 23 33 103 105 107
p5 26 24 20 106 112

=====
N(phi_i, alpha_k=a2, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 24 25 8 6 1
p2 35 43 31 7 7
p3 8 42 34 32 7
p4 11 17 34 24 38
p5 13 12 8 39 40

=====
N(phi_i, alpha_k=a3, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 11 12 3 2 4
p2 17 10 15 1 3
p3 3 11 19 10 3
p4 4 3 13 15 17
p5 7 5 3 16 16

=====
N(alpha_k, beta) -- raw counts
=====
beta 0 1
alpha
a1 1627 418
a2 339 207
a3 108 115

=====
Policy P(alpha_k | phi_i)
=====
alpha a1 a2 a3
phi_src
p1 0.824818 0.116788 0.058394
p2 0.761299 0.173729 0.064972
p3 0.700355 0.218085 0.081560
p4 0.678245 0.226691 0.095064
p5 0.644295 0.250559 0.105145

=====
1618 Transition P(phi_j | phi_i, alpha_k=a1)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.400442 0.384956 0.070796 0.068584 0.075221
p2 0.296846 0.296846 0.289425 0.068646 0.048237
p3 0.065823 0.331646 0.258228 0.281013 0.063291
p4 0.061995 0.088949 0.277628 0.283019 0.288410
p5 0.090278 0.083333 0.069444 0.368056 0.388889

```

```

=====
Transition P(phi_j | phi_i, alpha_k=a2)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.375000 0.390625 0.125000 0.093750 0.015625
p2 0.284553 0.349593 0.252033 0.056911 0.056911
p3 0.065041 0.341463 0.276423 0.260163 0.056911
p4 0.088710 0.137097 0.274194 0.193548 0.306452
p5 0.116071 0.107143 0.071429 0.348214 0.357143

=====
Transition P(phi_j | phi_i, alpha_k=a3)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.343750 0.375000 0.093750 0.062500 0.125000
p2 0.369565 0.217391 0.326087 0.021739 0.065217
p3 0.065217 0.239130 0.413043 0.217391 0.065217
p4 0.076923 0.057692 0.250000 0.288462 0.326923
p5 0.148936 0.106383 0.063830 0.340426 0.340426

=====
Penalty probabilities c_k = P(beta=1 | alpha_k)
=====
c_k empirical_penalty
alpha
a1 0.204401 0.204401
a2 0.379121 0.379121
a3 0.515695 0.515695

=====
Consistency checks
=====
[PASS] sum_k P(alpha_k | phi_i) == 1
[PASS] sum_j P(phi_j | phi_i, alpha_k) == 1
[PASS] c_k matches empirical frequencies

=====
Task 2 - Optimal Action Identification (source: estimation-result)
=====
Penalty probabilities c_k = P(beta = 1 | alpha_k):
alpha
a1 0.204401
a2 0.379121
a3 0.515695

alpha* = a1
c_* = 0.204401

=====
Ranking by c_k (ascending)
=====
c_k rank is_optimal
alpha
a1 0.204401 1 True
a2 0.379121 2 False
a3 0.515695 3 False

=====
Task 3 - e-optimality (source: estimation-result + optimal-result)
=====
a* = a1
min_i P(a*|p_i) over visited states = 0.644295
e = 1 - min_i P(a*|p_i) = 0.355705
Visited states (count): 5
Argmin state(s) (ties): ['p5']

P(a*|p_i) on visited states:
phi_src
p1 0.824818
p2 0.761299

```

1668

p3 0.700355
p4 0.678245
p5 0.644295

=====
Interpretation (assignment prompts)
=====

* Is the system strongly optimal?
Weakly - e is large: at least one visited state rarely chooses a* under the empirical policy.
* Is it noisy?
Yes - high e and/or high spread across states suggests heterogeneous or stochastic-looking action choices.
* Is behavior consistent across states?
Partially - some variation across states but not extreme.

Diagnostics: spread(max-min P)=0.180522, std(P)=0.071653

=====
Task 4 - Simulation (source: estimation-result, seed=42)
=====

Real rows: 3518 | Synthetic rows: 3518

|P_data - P_sim| - action distribution
alpha
a1 0.009949
a2 0.009949
a3 0.000000
L1 sum of abs diffs (actions): 0.019898

|P_data - P_sim| - phi_src visitation
phi_src
p1 0.009096
p2 0.003695
p3 0.005401
p4 0.007675
p5 0.000284
L1 sum: 0.026151

|P_data - P_sim| - phi_dst visitation
phi_dst
p1 0.010517
p2 0.002843
p3 0.005969
p4 0.007959
p5 0.000569
L1 sum: 0.027857

|mean_beta_data - mean_beta_sim| per action (penalty rate)
alpha
a1 0.005797
a2 0.003629
a3 0.024648
L1 sum (per-action |D|): 0.034074

Global penalty rate |D|: data=0.260659 sim=0.257248 |D|=0.003411

=====
main: Engin-vsa_dataset_full.csv (trial 6, removed 704) - Tasks 1--4 finished
=====

rows processed : 2814
optimal action : alpha* = a1 (c* = 0.204401)
epsilon (Task 3) : e = 0.355705 (min P(a*|p) = 0.644295)
simulation L1 : actions=0.0199 phi_src=0.0262 phi_dst=0.0279

=====
Task 1 - Estimation (dataset: Engin-vsa_dataset_full.csv (trial 7, removed 704))
=====

Rows in dataset : 2814
States observed : ['p1', 'p2', 'p3', 'p4', 'p5']
Actions observed: ['a1', 'a2', 'a3']
=====

```

N(phi_i, alpha_k) -- raw counts
=====
alpha a1 a2 a3
phi_src
p1 463 63 30
p2 546 124 44
p3 404 127 42
p4 361 125 57
p5 274 104 50

=====
N(phi_i, alpha_k=a1, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 180 183 32 31 37
p2 165 153 162 37 29
p3 27 143 101 109 24
p4 25 35 103 101 97
p5 20 32 18 108 96

=====
N(phi_i, alpha_k=a2, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 20 29 3 9 2
p2 34 42 32 8 8
p3 7 41 36 36 7
p4 12 16 32 29 36
p5 11 10 8 38 37

=====
N(phi_i, alpha_k=a3, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 11 9 3 3 4
p2 14 11 15 1 3
p3 2 13 16 10 1
p4 4 3 16 17 17
p5 7 7 3 19 14

=====
N(alpha_k, beta) -- raw counts
=====
beta 0 1
alpha
a1 1633 415
a2 338 205
a3 105 118

=====
Policy P(alpha_k | phi_i)
=====
alpha a1 a2 a3
phi_src
p1 0.832734 0.113309 0.053957
p2 0.764706 0.173669 0.061625
p3 0.705061 0.221640 0.073298
p4 0.664825 0.230203 0.104972
p5 0.640187 0.242991 0.116822

=====
Transition P(phi_j | phi_i, alpha_k=a1)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.388769 0.395248 0.069114 0.066955 0.079914
p2 0.302198 0.280220 0.296703 0.067766 0.053114
p3 0.066832 0.353960 0.250000 0.269802 0.059406
p4 0.069252 0.096953 0.285319 0.279778 0.268698
p5 0.072993 0.116788 0.065693 0.394161 0.350365

```

1818

1868

```

=====
Transition P(phi_j | phi_i, alpha_k=a2)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.317460 0.460317 0.047619 0.142857 0.031746
p2 0.274194 0.338710 0.258065 0.064516 0.064516
p3 0.055118 0.322835 0.283465 0.283465 0.055118
p4 0.096000 0.128000 0.256000 0.232000 0.288000
p5 0.105769 0.096154 0.076923 0.365385 0.355769

=====
Transition P(phi_j | phi_i, alpha_k=a3)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.366667 0.300000 0.100000 0.100000 0.133333
p2 0.318182 0.250000 0.340909 0.022727 0.068182
p3 0.047619 0.309524 0.380952 0.238095 0.023810
p4 0.070175 0.052632 0.280702 0.298246 0.298246
p5 0.140000 0.140000 0.060000 0.380000 0.280000

=====
Penalty probabilities c_k = P(beta=1 | alpha_k)
=====
c_k empirical_penalty
alpha
a1 0.202637 0.202637
a2 0.377532 0.377532
a3 0.529148 0.529148

=====
Consistency checks
=====
[PASS] sum_k P(alpha_k | phi_i) == 1
[PASS] sum_j P(phi_j | phi_i, alpha_k) == 1
[PASS] c_k matches empirical frequencies

=====
Task 2 - Optimal Action Identification (source: estimation-result)
=====
Penalty probabilities c_k = P(beta = 1 | alpha_k):
alpha
a1 0.202637
a2 0.377532
a3 0.529148

alpha* = a1
c_* = 0.202637

=====
Ranking by c_k (ascending)
=====
c_k rank is_optimal
alpha
a1 0.202637 1 True
a2 0.377532 2 False
a3 0.529148 3 False

=====
Task 3 - e-optimality (source: estimation-result + optimal-result)
=====
a* = a1
min_i P(a*|p_i) over visited states = 0.640187
e = 1 - min_i P(a*|p_i) = 0.359813
Visited states (count): 5
Argmin state(s) (ties): ['p5']

P(a*|p_i) on visited states:
phi_src
p1 0.832734
p2 0.764706

```

```

1918 p3 0.705061
      p4 0.664825
      p5 0.640187

=====
Interpretation (assignment prompts)
=====
* Is the system strongly optimal?
  Weakly - e is large: at least one visited state rarely chooses a* under the empirical policy.
* Is it noisy?
  Yes - high e and/or high spread across states suggests heterogeneous or stochastic-looking action
  choices.
* Is behavior consistent across states?
  Partially - some variation across states but not extreme.

Diagnostics: spread(max-min P)=0.192547, std(P)=0.077996

=====
Task 4 - Simulation (source: estimation-result, seed=42)
=====
Real rows: 3518 | Synthetic rows: 3518

|P_data - P_sim| - action distribution
alpha
a1 0.008528
a2 0.008243
a3 0.000284
  L1 sum of abs diffs (actions): 0.017055

|P_data - P_sim| - phi_src visitation
phi_src
p1 0.005117
p2 0.010802
p3 0.000853
p4 0.007391
p5 0.007675
  L1 sum: 0.031836

|P_data - P_sim| - phi_dst visitation
phi_dst
p1 0.006822
p2 0.009096
p3 0.000569
p4 0.007391
p5 0.007959
  L1 sum: 0.031836

|mean_beta_data - mean_beta_sim| per action (penalty rate)
alpha
a1 0.006585
a2 0.006006
a3 0.033631
  L1 sum (per-action |D|): 0.046222

Global penalty rate |D|: data=0.260659 sim=0.255827 |D|=0.004832

=====
main: Engin-vsa_dataset_full.csv (trial 7, removed 704) - Tasks 1--4 finished
=====
  rows processed : 2814
  optimal action : alpha* = a1 (c* = 0.202637)
  epsilon (Task 3) : e = 0.359813 (min P(a*|p) = 0.640187)
  simulation L1 : actions=0.0171 phi_src=0.0318 phi_dst=0.0318

=====
Task 1 - Estimation (dataset: Engin-vsa_dataset_full.csv (trial 8, removed 704))
=====
Rows in dataset : 2814
States observed : ['p1', 'p2', 'p3', 'p4', 'p5']
Actions observed: ['a1', 'a2', 'a3']

=====

```

1968

```

N(phi_i, alpha_k) -- raw counts
=====
alpha a1 a2 a3
phi_src
p1 445 58 31
p2 545 130 47
p3 402 130 49
p4 375 126 53
p5 265 111 47

=====
N(phi_i, alpha_k=a1, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 178 174 31 29 33
p2 160 159 160 39 27
p3 23 133 112 106 28
p4 24 31 106 102 112
p5 22 26 17 107 93

=====
N(phi_i, alpha_k=a2, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 20 22 7 7 2
p2 38 43 35 7 7
p3 8 41 36 38 7
p4 10 16 38 28 34
p5 15 12 8 38 38

=====
N(phi_i, alpha_k=a3, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 13 9 3 4 2
p2 17 14 13 0 3
p3 4 13 16 13 3
p4 4 3 17 14 15
p5 5 5 3 19 15

=====
N(alpha_k, beta) -- raw counts
=====
beta 0 1
alpha
a1 1632 400
a2 343 212
a3 109 118

=====
Policy P(alpha_k | phi_i)
=====
alpha a1 a2 a3
phi_src
p1 0.833333 0.108614 0.058052
p2 0.754848 0.180055 0.065097
p3 0.691910 0.223752 0.084337
p4 0.676895 0.227437 0.095668
p5 0.626478 0.262411 0.111111

=====
Transition P(phi_j | phi_i, alpha_k=a1)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.400000 0.391011 0.069663 0.065169 0.074157
p2 0.293578 0.291743 0.293578 0.071560 0.049541
p3 0.057214 0.330846 0.278607 0.263682 0.069652
p4 0.064000 0.082667 0.282667 0.272000 0.298667
p5 0.083019 0.098113 0.064151 0.403774 0.350943

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```

=====
Transition P(phi_j | phi_i, alpha_k=a2)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.344828 0.379310 0.120690 0.120690 0.034483
p2 0.292308 0.330769 0.269231 0.053846 0.053846
p3 0.061538 0.315385 0.276923 0.292308 0.053846
p4 0.079365 0.126984 0.301587 0.222222 0.269841
p5 0.135135 0.108108 0.072072 0.342342 0.342342

=====
Transition P(phi_j | phi_i, alpha_k=a3)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.419355 0.290323 0.096774 0.129032 0.064516
p2 0.361702 0.297872 0.276596 0.000000 0.063830
p3 0.081633 0.265306 0.326531 0.265306 0.061224
p4 0.075472 0.056604 0.320755 0.264151 0.283019
p5 0.106383 0.106383 0.063830 0.404255 0.319149

=====
Penalty probabilities c_k = P(beta=1 | alpha_k)
=====
c_k empirical_penalty
alpha
a1 0.196850 0.196850
a2 0.381982 0.381982
a3 0.519824 0.519824

=====
Consistency checks
=====
[PASS] sum_k P(alpha_k | phi_i) == 1
[PASS] sum_j P(phi_j | phi_i, alpha_k) == 1
[PASS] c_k matches empirical frequencies

=====
Task 2 - Optimal Action Identification (source: estimation-result)
=====
Penalty probabilities c_k = P(beta = 1 | alpha_k):
alpha
a1 0.196850
a2 0.381982
a3 0.519824

alpha* = a1
c_* = 0.196850

=====
Ranking by c_k (ascending)
=====
c_k rank is_optimal
alpha
a1 0.196850 1 True
a2 0.381982 2 False
a3 0.519824 3 False

=====
Task 3 - e-optimality (source: estimation-result + optimal-result)
=====
a* = a1
min_i P(a*|p_i) over visited states = 0.626478
e = 1 - min_i P(a*|p_i) = 0.373522
Visited states (count): 5
Argmin state(s) (ties): ['p5']

P(a*|p_i) on visited states:
phi_src
p1 0.833333
p2 0.754848

```

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```

p3 0.691910
p4 0.676895
p5 0.626478

=====
Interpretation (assignment prompts)
=====
* Is the system strongly optimal?
  Weakly - e is large: at least one visited state rarely chooses a* under the empirical policy.
* Is it noisy?
  Yes - high e and/or high spread across states suggests heterogeneous or stochastic-looking action
  choices.
* Is behavior consistent across states?
  No - behavior differs markedly across visited states (large spread in P(a*|p)).

Diagnostics: spread(max-min P)=0.206856, std(P)=0.079683

=====
Task 4 - Simulation (source: estimation-result, seed=42)
=====
Real rows: 3518 | Synthetic rows: 3518

|P_data - P_sim| - action distribution
alpha
a1 0.003411
a2 0.004832
a3 0.001421
  L1 sum of abs diffs (actions): 0.009665

|P_data - P_sim| - phi_src visitation
phi_src
p1 0.004548
p2 0.002843
p3 0.003695
p4 0.001137
p5 0.004264
  L1 sum: 0.016487

|P_data - P_sim| - phi_dst visitation
phi_dst
p1 0.005969
p2 0.004264
p3 0.003411
p4 0.000853
p5 0.004264
  L1 sum: 0.018761

|mean_beta_data - mean_beta_sim| per action (penalty rate)
alpha
a1 0.014596
a2 0.005228
a3 0.018727
  L1 sum (per-action |D|): 0.038550

Global penalty rate |D|: data=0.260659 sim=0.252132 |D|=0.008528

=====
main: Engin-vsa_dataset_full.csv (trial 8, removed 704) - Tasks 1--4 finished
=====
  rows processed : 2814
  optimal action : alpha* = a1 (c* = 0.196850)
  epsilon (Task 3) : e = 0.373522 (min P(a*|p) = 0.626478)
  simulation L1 : actions=0.0097 phi_src=0.0165 phi_dst=0.0188

=====
Task 1 - Estimation (dataset: Engin-vsa_dataset_full.csv (trial 9, removed 704))
=====
Rows in dataset : 2814
States observed : ['p1', 'p2', 'p3', 'p4', 'p5']
Actions observed: ['a1', 'a2', 'a3']
=====

```

```

N(phi_i, alpha_k) -- raw counts
=====
alpha a1 a2 a3
phi_src
p1 456 63 28
p2 537 122 47
p3 419 125 47
p4 358 121 49
p5 279 112 51

2218 =====
N(phi_i, alpha_k=a1, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 184 177 33 29 33
p2 157 167 149 36 28
p3 25 135 114 113 32
p4 22 31 101 105 99
p5 26 27 16 106 104

=====
N(phi_i, alpha_k=a2, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 19 27 8 7 2
p2 33 46 32 5 6
p3 7 43 30 40 5
p4 11 11 36 25 38
p5 14 11 10 39 38

=====
N(phi_i, alpha_k=a3, phi_j) -- raw counts
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 13 6 1 4 4
p2 15 13 15 1 3
p3 3 14 15 13 2
p4 5 2 15 14 13
p5 5 4 2 21 19

=====
N(alpha_k, beta) -- raw counts
=====
beta 0 1
alpha
a1 1642 407
a2 341 202
a3 108 114

=====
Policy P(alpha_k | phi_i)
=====
alpha a1 a2 a3
phi_src
p1 0.833638 0.115174 0.051188
p2 0.760623 0.172805 0.066572
p3 0.708968 0.211506 0.079526
2268 p4 0.678030 0.229167 0.092803
p5 0.631222 0.253394 0.115385

=====
Transition P(phi_j | phi_i, alpha_k=a1)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.403509 0.388158 0.072368 0.063596 0.072368
p2 0.292365 0.310987 0.277467 0.067039 0.052142
p3 0.059666 0.322196 0.272076 0.269690 0.076372
p4 0.061453 0.086592 0.282123 0.293296 0.276536
p5 0.093190 0.096774 0.057348 0.379928 0.372760

```

2318

```

=====
Transition P(phi_j | phi_i, alpha_k=a2)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.301587 0.428571 0.126984 0.111111 0.031746
p2 0.270492 0.377049 0.262295 0.040984 0.049180
p3 0.056000 0.344000 0.240000 0.320000 0.040000
p4 0.090909 0.090909 0.297521 0.206612 0.314050
p5 0.125000 0.098214 0.089286 0.348214 0.339286

=====
Transition P(phi_j | phi_i, alpha_k=a3)
=====
phi_dst p1 p2 p3 p4 p5
phi_src
p1 0.464286 0.214286 0.035714 0.142857 0.142857
p2 0.319149 0.276596 0.319149 0.021277 0.063830
p3 0.063830 0.297872 0.319149 0.276596 0.042553
p4 0.102041 0.040816 0.306122 0.285714 0.265306
p5 0.098039 0.078431 0.039216 0.411765 0.372549

=====
Penalty probabilities c_k = P(beta=1 | alpha_k)
=====
c_k empirical_penalty
alpha
a1 0.198633 0.198633
a2 0.372007 0.372007
a3 0.513514 0.513514

=====
Consistency checks
=====
[PASS] sum_k P(alpha_k | phi_i) == 1
[PASS] sum_j P(phi_j | phi_i, alpha_k) == 1
[PASS] c_k matches empirical frequencies

=====
Task 2 - Optimal Action Identification (source: estimation-result)
=====
Penalty probabilities c_k = P(beta = 1 | alpha_k):
alpha
a1 0.198633
a2 0.372007
a3 0.513514

alpha* = a1
c_* = 0.198633

=====
Ranking by c_k (ascending)
=====
c_k rank is_optimal
alpha
a1 0.198633 1 True
a2 0.372007 2 False
a3 0.513514 3 False

=====
Task 3 - e-optimality (source: estimation-result + optimal-result)
=====
a* = a1
min_i P(a*|p_i) over visited states = 0.631222
e = 1 - min_i P(a*|p_i) = 0.368778
Visited states (count): 5
Argmin state(s) (ties): ['p5']

P(a*|p_i) on visited states:
phi_src
p1 0.833638
p2 0.760623

```

```

p3 0.708968
p4 0.678030
p5 0.631222

=====
Interpretation (assignment prompts)
=====
* Is the system strongly optimal?
  Weakly - e is large: at least one visited state rarely chooses a* under the empirical policy.
* Is it noisy?
  Yes - high e and/or high spread across states suggests heterogeneous or stochastic-looking action
  choices.
* Is behavior consistent across states?
  No - behavior differs markedly across visited states (large spread in  $P(a|p)$ ).

Diagnostics: spread(max-min P)=0.202416, std(P)=0.077938

=====
Task 4 - Simulation (source: estimation-result, seed=42)
=====
Real rows: 3518 | Synthetic rows: 3518

|P_data - P_sim| - action distribution
alpha
a1 0.010517
a2 0.009096
a3 0.001421
  L1 sum of abs diffs (actions): 0.021035

|P_data - P_sim| - phi_src visitation
phi_src
p1 0.003411
p2 0.006254
p3 0.009096
p4 0.001421
p5 0.001990
  L1 sum: 0.022172

|P_data - P_sim| - phi_dst visitation
phi_dst
p1 0.005117
p2 0.004832
p3 0.009380
p4 0.001137
p5 0.001706
  L1 sum: 0.022172

|mean_beta_data - mean_beta_sim| per action (penalty rate)
alpha
a1 0.013678
a2 0.001211
a3 0.016445
  L1 sum (per-action |D|): 0.031334

Global penalty rate |D|: data=0.260659 sim=0.249574 |D|=0.011086

=====
main: Engin-vsa_dataset_full.csv (trial 9, removed 704) - Tasks 1--4 finished
=====
  rows processed : 2814
  optimal action : alpha* = a1 (c* = 0.198633)
  epsilon (Task 3) : e = 0.368778 (min  $P(a|p)$  = 0.631222)
  simulation L1 : actions=0.0210 phi_src=0.0222 phi_dst=0.0222

=====
Task 5 - Perturbation (20% rows removed, 10 trial(s))
=====
Dataset : D:\Documents\00-PhD\04-CSE8003\assignments\term_project\Engin-vsa_dataset_full.csv
Rows : 3518 full

Baseline (full data, Tasks 1--4)
  alpha* = a1 c* = 0.199608

```

epsilon = 0.370166

Perturbed runs

Engin-vsa_dataset_full.csv (trial 0, removed 704): alpha*=a1 epsilon=0.391705 L1_act=0.0057
Engin-vsa_dataset_full.csv (trial 1, removed 704): alpha*=a1 epsilon=0.374408 L1_act=0.0176
Engin-vsa_dataset_full.csv (trial 2, removed 704): alpha*=a1 epsilon=0.374134 L1_act=0.0028
Engin-vsa_dataset_full.csv (trial 3, removed 704): alpha*=a1 epsilon=0.351039 L1_act=0.0142
Engin-vsa_dataset_full.csv (trial 4, removed 704): alpha*=a1 epsilon=0.370968 L1_act=0.0165
Engin-vsa_dataset_full.csv (trial 5, removed 704): alpha*=a1 epsilon=0.394366 L1_act=0.0085
Engin-vsa_dataset_full.csv (trial 6, removed 704): alpha*=a1 epsilon=0.355705 L1_act=0.0199
Engin-vsa_dataset_full.csv (trial 7, removed 704): alpha*=a1 epsilon=0.359813 L1_act=0.0171
Engin-vsa_dataset_full.csv (trial 8, removed 704): alpha*=a1 epsilon=0.373522 L1_act=0.0097
Engin-vsa_dataset_full.csv (trial 9, removed 704): alpha*=a1 epsilon=0.368778 L1_act=0.0210

epsilon mean=0.371444 std=0.013996 delta_mean_vs_baseline=+0.001278

Spread across states per action (max_phi c_k - min_phi c_k):

Baseline: a1=0.051272 a2=0.099706 a3=0.243992

Perturbed: a1=mean 0.050590 std 0.010036 a2=mean 0.120915 std 0.018252 a3=mean 0.249805 std 0.040117

|c_k_per_state - baseline| max cell delta:

mean=0.051845 std=0.021050 max=0.094017

Task 6 - Model assumption check

c_k(phi_i) = P(beta=1 | alpha_k, phi_i)

Engin-vsa_dataset_full.csv (baseline) (n=3518 rows):

alpha a1 a2 a3

phi_src

p1 0.205934 0.346154 0.461538

p2 0.225564 0.350318 0.362069

p3 0.195695 0.445860 0.517241

p4 0.174292 0.363636 0.606061

p5 0.178363 0.384058 0.603175

Engin-vsa_dataset_full.csv (trial 0, removed 704) (n=2814 rows):

alpha a1 a2 a3

phi_src

p1 0.207506 0.358209 0.451613

p2 0.224953 0.351562 0.360000

p3 0.189526 0.459854 0.555556

p4 0.183333 0.359375 0.568627

p5 0.193182 0.398305 0.634615

Engin-vsa_dataset_full.csv (trial 1, removed 704) (n=2814 rows):

alpha a1 a2 a3

phi_src

p1 0.191964 0.322581 0.448276

p2 0.225564 0.373134 0.390244

p3 0.195181 0.436508 0.488889

p4 0.183024 0.385246 0.606557

p5 0.185606 0.379630 0.620000

Engin-vsa_dataset_full.csv (trial 2, removed 704) (n=2814 rows):

alpha a1 a2 a3

phi_src

p1 0.201285 0.380952 0.483871

p2 0.226236 0.344000 0.314815

p3 0.203474 0.451128 0.489362

p4 0.188732 0.351145 0.608696

p5 0.177122 0.357798 0.622642

Engin-vsa_dataset_full.csv (trial 3, removed 704) (n=2814 rows):

alpha a1 a2 a3

phi_src

p1 0.219089 0.367647 0.382353

p2 0.219282 0.358779 0.377778

p3 0.214112 0.440000 0.567568

p4 0.175000 0.333333 0.596491

p5 0.174377 0.417476 0.551020

Engin-vsa_dataset_full.csv (trial 4, removed 704) (n=2814 rows):

2468

2518

```
alpha a1 a2 a3
phi_src
p1 0.207589 0.323077 0.472222
p2 0.226415 0.312500 0.347826
p3 0.177778 0.467213 0.500000
p4 0.177249 0.400000 0.666667
p5 0.183150 0.407407 0.660377

Engin-vsa_dataset_full.csv (trial 5, removed 704) (n=2814 rows):
alpha a1 a2 a3
phi_src
p1 0.208155 0.333333 0.555556
p2 0.222628 0.338583 0.408163
p3 0.195704 0.443548 0.500000
p4 0.161383 0.336066 0.618182
p5 0.178295 0.379310 0.538462

Engin-vsa_dataset_full.csv (trial 6, removed 704) (n=2814 rows):
alpha a1 a2 a3
phi_src
p1 0.221239 0.375000 0.437500
p2 0.233766 0.341463 0.391304
p3 0.197468 0.447154 0.521739
p4 0.164420 0.354839 0.596154
p5 0.184028 0.375000 0.595745

Engin-vsa_dataset_full.csv (trial 7, removed 704) (n=2814 rows):
alpha a1 a2 a3
phi_src
p1 0.205184 0.333333 0.466667
p2 0.230769 0.306452 0.386364
p3 0.193069 0.425197 0.500000
p4 0.171745 0.384000 0.614035
p5 0.197080 0.423077 0.620000

Engin-vsa_dataset_full.csv (trial 8, removed 704) (n=2814 rows):
alpha a1 a2 a3
phi_src
p1 0.200000 0.293103 0.419355
p2 0.229358 0.353846 0.382979
p3 0.179104 0.438462 0.530612
p4 0.178667 0.357143 0.603774
p5 0.177358 0.423423 0.617021

Engin-vsa_dataset_full.csv (trial 9, removed 704) (n=2814 rows):
alpha a1 a2 a3
phi_src
p1 0.201754 0.301587 0.428571
p2 0.219739 0.344262 0.361702
p3 0.190931 0.440000 0.510638
p4 0.184358 0.371901 0.591837
p5 0.182796 0.366071 0.627451

Process finished with exit code 0
```